

Posterior Variance Collapse in Variational Inference: Diagnosis and Laplace Correction Across Three Bayesian Regression Models

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Abstract

Mean-field variational inference turns intractable posterior computation into an optimisation problem by maximising the Evidence Lower Bound (ELBO), but the mean-field factorisation systematically underestimates posterior uncertainty — a problem known as *posterior variance collapse*. This report diagnoses that phenomenon across three Bayesian regression models and proposes a Laplace approximation correction to recover more faithful credible intervals.

Five case studies were conducted (linear, quadratic, logistic, hierarchical linear, and hierarchical logistic regression). The quadratic (CS2) and logistic (CS3) cases are excluded from the correction analysis because their collapse is negligible ($\sigma_{\text{VB}}/\sigma_{\text{Gibbs}} > 0.97$ for all parameters), so the focus falls on CS1 (Bayesian linear regression), CS4 (hierarchical linear), and CS5 (hierarchical logistic).

In each retained case, four optimisation methods — CAVI, gradient ascent, Newton, and BFGS — are used to maximise the ELBO. Once CAVI has converged, its posterior mean $\hat{\boldsymbol{\mu}}$ is accepted as the best available point estimate, and the covariance is corrected by inverting the Hessian of the joint log-posterior evaluated at that point:

$$\hat{\boldsymbol{\Sigma}}_{\text{corr}} = \left(-\nabla^2 \log p(\boldsymbol{\theta} \mid \mathbf{y}) \right)^{-1} \Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\mu}}}.$$

The Hessian is computed by central finite differences at a single point; CAVI is *not* re-run, so the additional cost is negligible. Corrected credible intervals are then compared against the Gibbs gold standard.

The correction is effective in both hierarchical settings. For CS4 (hierarchical linear), the intercept SD ratio rises from 0.130 to 0.764 while the slope already reaches 0.998. For CS5 (hierarchical logistic), both coefficients move from ≈ 0.46 to 0.82–0.88. In CS1 the collapse is mild to begin with (≈ 0.98) and the correction changes nothing, which is reassuring: it confirms that the problem is structural rather than a numerical artefact.

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Introduction

Bayesian inference and the intractability problem

Bayesian regression places a prior distribution on parameters $\boldsymbol{\theta}$ and updates it via the likelihood to obtain the posterior:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}}. \quad (1)$$

The denominator of (1) is analytically tractable only for conjugate likelihood–prior pairs. For all three models studied here the integral is intractable, though for different reasons: hierarchical models introduce high-dimensional latent variables, and the logistic likelihood has no conjugate prior.

The three models studied are shown below in parallel, with each difference highlighted:

$$\text{CS1 (linear): } y_i \mid \boldsymbol{\beta}, \tau_e \sim \mathcal{N}\left(\underbrace{\mathbf{x}_i^T \boldsymbol{\beta}}_{\text{fixed effects}}, \tau_e^{-1}\right), \quad p(\boldsymbol{\beta}, \tau_e) \propto 1 \text{ (flat)},$$

$$\text{CS4 (hier. linear): } y_{ij} \mid \boldsymbol{\beta}, u_j, \tau_e \sim \mathcal{N}\left(\underbrace{\mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j}_{\text{fixed + random}}, \tau_e^{-1}\right), \quad u_j \sim \mathcal{N}(0, \tau_u^{-1}), \quad \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}),$$

$$\text{CS5 (hier. logistic): } y_{ij} \mid \boldsymbol{\beta}, u_j \sim \underbrace{\text{Bernoulli}}_{\text{non-Gaussian}}\left(\sigma\left(\underbrace{\mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j}_{\text{fixed + random}}\right)\right), \quad u_j \sim \mathcal{N}(0, \tau_u^{-1}), \quad \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}).$$

In all three models $\boldsymbol{\beta} \in \mathbb{R}^2$ with $\mathbf{x}_i = (1, x_i)^T$, and the group index $j \in \{1, \dots, 5\}$ applies to CS4 and CS5. CS4 and CS5 share the same random-effect structure; they differ in whether the observation model is Gaussian (CS4, with additional noise precision τ_e) or Bernoulli-logistic (CS5, with no explicit noise variance).

Scope of this report

A total of five case studies were conducted, covering linear, quadratic, logistic, hierarchical linear, and hierarchical logistic regression. **Case Studies 2 (quadratic) and 3 (logistic) are excluded from the correction analysis** because their posterior SD ratios $\sigma_{\text{VB}}/\sigma_{\text{Gibbs}}$ are greater than 0.97 for all parameters — collapse is negligible and no correction is warranted. The analysis focuses on CS1, CS4, and CS5, where collapse ranges from mild to severe.

Variational Inference as optimisation

Variational Inference (VI) replaces (1) with the optimisation problem

$$\max_{q \in \mathcal{Q}} \mathcal{L}(q) = \mathbb{E}_q \left[\log \frac{p(\mathbf{y}, \boldsymbol{\theta})}{q(\boldsymbol{\theta})} \right] = \mathbb{E}_q [\log p(\mathbf{y} \mid \boldsymbol{\theta})] - \text{KL}(q \parallel p), \quad (2)$$

where $\mathcal{L}(q)$ is the Evidence Lower Bound (ELBO). Maximising (2) is equivalent to minimising the KL divergence between the approximation q and the true posterior. In the mean-field approximation the posterior is factorised as $q(\boldsymbol{\theta}) = q(\boldsymbol{\beta}) q(\tau_e) \prod_j q(u_j) q(\tau_u)$, and each factor is optimised in turn (CAVI) or jointly via gradient-based methods.

Four optimisation methods are applied to the ELBO in each case study: CAVI, gradient ascent with Armijo backtracking, Newton’s method with finite-difference Hessian, and BFGS with

Wolfe conditions. Reference posteriors are obtained from Gibbs samplers (conjugate for linear models; Pólya–Gamma (Polson2013) for logistic models).

The variance-collapse problem and the Laplace correction

A known failure mode of mean-field VI is *posterior variance collapse*: the optimised approximation q^* systematically underestimates posterior uncertainty because the mean-field factorisation eliminates posterior correlations and reduces effective parameter uncertainty. Concretely, the CAVI update for the β covariance in CS1 is

$$\Sigma_\beta = (\mathbb{E}[\tau_e] \mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1},$$

which treats τ_e as fixed at its variational mean rather than integrating over its uncertainty; this systematically tightens the covariance. In hierarchical models the collapse is amplified because the variational distribution over group effects u_j is also factorised from β .

The remedy proposed here is the **Laplace approximation** centred at the CAVI solution. The CAVI mean $\hat{\mu}$ is accepted as the best available point estimate; the full joint log-posterior $\log p(\theta \mid \mathbf{y})$ is then written in unconstrained coordinates $(\beta, \mathbf{u}, \log \tau_e, \log \tau_u)$ and its Hessian $H = \nabla^2 \log p|_{\hat{\mu}}$ is evaluated by central finite differences at that single point. The corrected covariance is $\hat{\Sigma}_{\text{corr}} = (-H)^{-1}$. CAVI is *not* re-run: only the log-posterior curvature is needed, not the ELBO. The gradient-based methods (Newton, BFGS, gradient ascent) serve to cross-check the Hessian and confirm that $\hat{\mu}$ sits near the log-posterior mode.

Case Study 1 — Linear Regression ($n = 50, p = 2$): Serves as the baseline. With CAVI SD ratios of ≈ 0.98 , collapse is mild and the Laplace correction leaves the intervals essentially unchanged — confirming that the problem does not arise when the posterior is near-Gaussian.

Case Study 4 — Hierarchical Linear ($N = 150, J = 5, p = 2$): Shows severe asymmetric collapse: the intercept β_0 has an SD ratio of only 0.130 while the slope is barely affected (0.987), because β_0 is entangled with the group random effects in a way that β_1 is not. The Laplace correction lifts β_0 to 0.764 and β_1 to 0.998.

Case Study 5 — Hierarchical Logistic ($N = 150, J = 5, p = 2$): Exhibits symmetric collapse, with both coefficients stuck near 0.46 — a pattern that the Laplace correction recovers to 0.82–0.88.

1 Common Methodology

1.1 Variational Inference as Optimisation

The mean-field variational approximation factorises the joint posterior as $q(\beta, \tau_e) = q(\beta) q(\tau_e)$ for the Gaussian regression cases, or $q(\beta) = \mathcal{N}(\mu, \Sigma)$ for the logistic case. Maximising the ELBO over the variational parameters ϕ is equivalent to minimising the KL divergence $\text{KL}(q\|p)$ between the approximation and the true posterior.

1.2 Constraint Handling: Cholesky Reparameterisation

The covariance matrix Σ must remain symmetric positive definite at every iteration. The set of symmetric positive definite matrices has a curved boundary: a standard Euclidean gradient step can cross outside it, producing an invalid covariance. The solution is to parameterise over the Cholesky factor $\mathbf{L}$, where $\Sigma = \mathbf{L}\mathbf{L}^T$ and $\mathbf{L}$ is lower-triangular with positive diagonal entries. The diagonal entries are reparameterised as $L_{ii} = e^{\tilde{\ell}_{ii}}$ to allow unconstrained optimisation. This converts the constraint into a smooth bijection and ensures every gradient step produces a valid covariance.

1.3 Optimisation Methods

All four methods share the same objective (the ELBO), the same reparameterisation, and the same convergence criteria.

CAVI: Coordinate Ascent Variational Inference. Closed-form coordinate updates derived from the exponential family structure. Each update exactly maximises the ELBO over one block of parameters while holding the others fixed. Most efficient method; requires conjugate structure.

Gradient ascent: First-order method. Search direction is the gradient $\mathbf{d}^{(t)} = \nabla \mathcal{L}(\phi^{(t)})$. Step size determined by Armijo backtracking (sufficient increase condition).

Newton’s method: Second-order method. Search direction is $\mathbf{d}^{(t)} = -\mathbf{H}^{-1}\nabla \mathcal{L}$, where $\mathbf{H}$ is the Hessian approximated by finite differences (25 evaluations per iteration). Fast convergence near the optimum; expensive per iteration.

BFGS: Quasi-Newton method. Builds a curvature approximation from gradient differences, requiring only gradient evaluations. Step size determined by Wolfe conditions to maintain positive definiteness of the approximate inverse Hessian ($\mathbf{y}_t^T \mathbf{s}_t > 0$). Best practical compromise between convergence speed and cost.

For all line-search methods, the search direction must be an ascent direction: $\nabla \mathcal{L}(\phi^{(t)})^T \mathbf{d}^{(t)} > 0$.

1.4 Convergence Criteria

Iteration terminates when any of the following conditions is satisfied:

- (i) Gradient norm: $\|\nabla \mathcal{L}\| < \varepsilon_g$.
- (ii) Relative ELBO change: $|\Delta \mathcal{L}|/|\mathcal{L}| < \varepsilon_r$.
- (iii) Parameter change: $\|\Delta \phi\| < \varepsilon_\phi$.
- (iv) Maximum iterations: $t \geq t_{\max}$.

CAVI additionally monitors the maximum relative change in variational parameters.

1.5 Reference Samplers

Two types of Gibbs sampler are used as gold-standard references.

Conjugate Gibbs (linear and quadratic cases). The Normal–Gamma conjugate structure of the Bayesian regression model (Gaussian likelihood with Gaussian prior on β and Gamma prior on precision τ_e) admits closed-form Gibbs updates:

$$\beta \mid \tau_e, \mathbf{y} \sim \mathcal{N}(\mu_n, \Sigma_n), \quad (3)$$

$$\tau_e \mid \beta, \mathbf{y} \sim \text{Gamma}(a_n, b_n), \quad (4)$$

where $\Sigma_n = (\tau_e \mathbf{X}^T \mathbf{X} + \lambda_\beta \mathbf{I})^{-1}$ and $\mu_n = \tau_e \Sigma_n \mathbf{X}^T \mathbf{y}$.

Pólya–Gamma Gibbs (logistic case). The Pólya–Gamma (PG) sampler of Polson2013<empty citation> augments the logistic model with latent variables $\omega_i \sim \text{PG}(1, x_i^T \beta)$, rendering β conditionally Gaussian:

$$\beta \mid \omega, \mathbf{y} \sim \mathcal{N}(\Sigma_\omega (\mathbf{X}^T \kappa), \Sigma_\omega), \quad \Sigma_\omega = (\mathbf{X}^T \Omega \mathbf{X} + \Sigma_0^{-1})^{-1}, \quad (5)$$

where $\Omega = \text{diag}(\omega_i)$ and $\kappa_i = y_i - 1/2$. The PG draw requires T Gamma variates per latent variable; the truncation $T = 50$ is used throughout.

1.6 Common Notation

Symbol	Meaning
Bayesian model	
$\mathbf{X}$	Design matrix ($n \times p$), rows x_i^T
$\mathbf{y}$	Response vector
$\boldsymbol{\beta}$	Regression coefficient vector
τ_e	Observation-noise precision (Gaussian cases)
n, p	Number of observations; number of predictors
λ_β	Prior precision on $\boldsymbol{\beta}$
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian approximation
$\boldsymbol{\mu}, \boldsymbol{\Sigma}$	Mean and covariance of $q(\boldsymbol{\beta})$
$\mathbf{L}$	Cholesky factor, $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$
$\mathcal{L}(\phi)$	Evidence Lower Bound (ELBO)
Optimisation	
$\mathbf{d}^{(t)}$	Search direction at iteration t
α_t	Step size at iteration t
$\mathbf{s}_t, \mathbf{y}_t$	Parameter-change and gradient-difference vectors (BFGS)
Logistic-specific	
$\sigma(\cdot)$	Logistic sigmoid, $\sigma(\psi) = (1 + e^{-\psi})^{-1}$
ξ_i	Jaakkola–Jordan tightness parameter
ω_i	Pólya–Gamma latent variable
κ_i	Pseudo-response $\kappa_i = y_i - 1/2$
ESS	Effective sample size
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\theta)$	Cost function = $\mathcal{L}(\phi)$ (ELBO)
Φ	Regressor matrix = $\mathbf{X}$

2 Case Study 1: Bayesian Linear Regression

2.1 Model

The Bayesian linear regression model is

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n), \quad (6)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p), \quad \lambda_\beta = 0.1, \quad (7)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e). \quad (8)$$

The mean-field variational approximation is $q(\boldsymbol{\beta}, \tau_e) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \cdot \text{Gamma}(a_e, b_e)$.

The test case uses $n = 50$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)^T$ ($p = 2$). Data are generated with noise standard deviation $\sigma = 0.5$.

2.2 ELBO

Under the Normal–Gamma variational family, the ELBO is

$$\mathcal{L}(\phi) = \mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \tau_e)] + \mathbb{E}_q[\log p(\boldsymbol{\beta})] + \mathbb{E}_q[\log p(\tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta})] - \mathbb{E}_q[\log q(\tau_e)]. \quad (9)$$

Each term has a closed-form expression:

$$\mathbb{E}_q[\log p(\mathbf{y} | \boldsymbol{\beta}, \tau_e)] = \frac{n}{2}(\psi(a_e) - \log b_e) - \frac{a_e}{2b_e}(\|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}\boldsymbol{\Sigma}_\beta\mathbf{X}^T)) - \frac{n}{2}\log(2\pi), \quad (10)$$

$$\text{KL}(q(\boldsymbol{\beta})\|p(\boldsymbol{\beta})) = \frac{1}{2}[\lambda_\beta \text{tr}(\boldsymbol{\Sigma}_\beta) + \lambda_\beta\|\boldsymbol{\mu}_\beta\|^2 - p - \log \det(\lambda_\beta\boldsymbol{\Sigma}_\beta)], \quad (11)$$

with analogous terms for the Gamma factor. Full derivations are in the individual linear report (Appendix ??).

2.3 Results

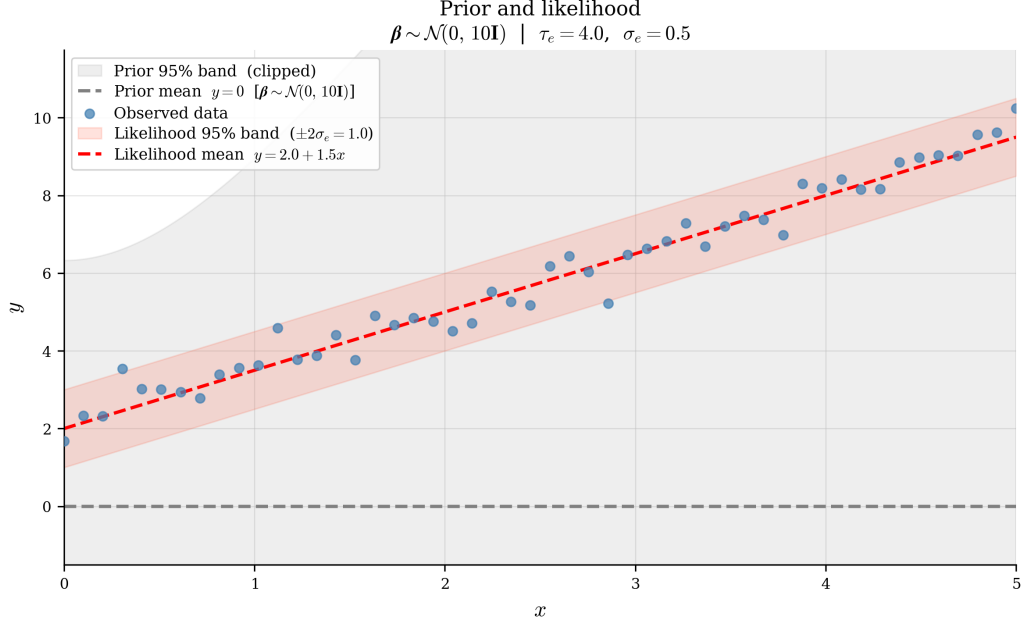


Figure 1: Synthetic data for Case Study 1: $n = 50$ observations, $x_i \sim \text{Uniform}(-2, 2)$, true $\boldsymbol{\beta} = (1.0, 2.0)$, noise $\sigma = 0.5$.

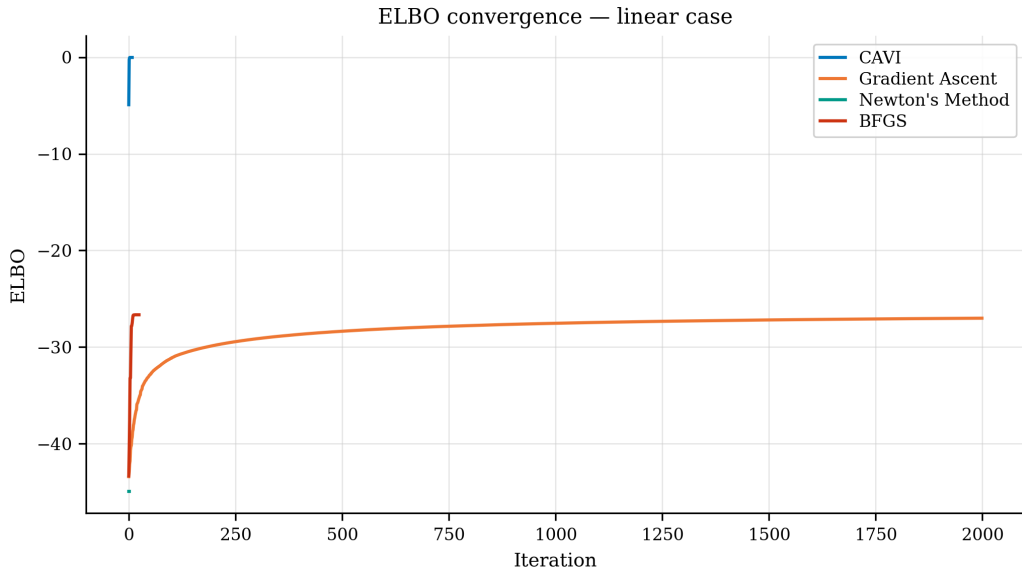


Figure 2: ELBO convergence for all four optimisation methods on the linear test case. CAVI converges in approximately 50 iterations; gradient ascent requires 2000. BFGS matches CAVI's final ELBO in fewer than 50 iterations.

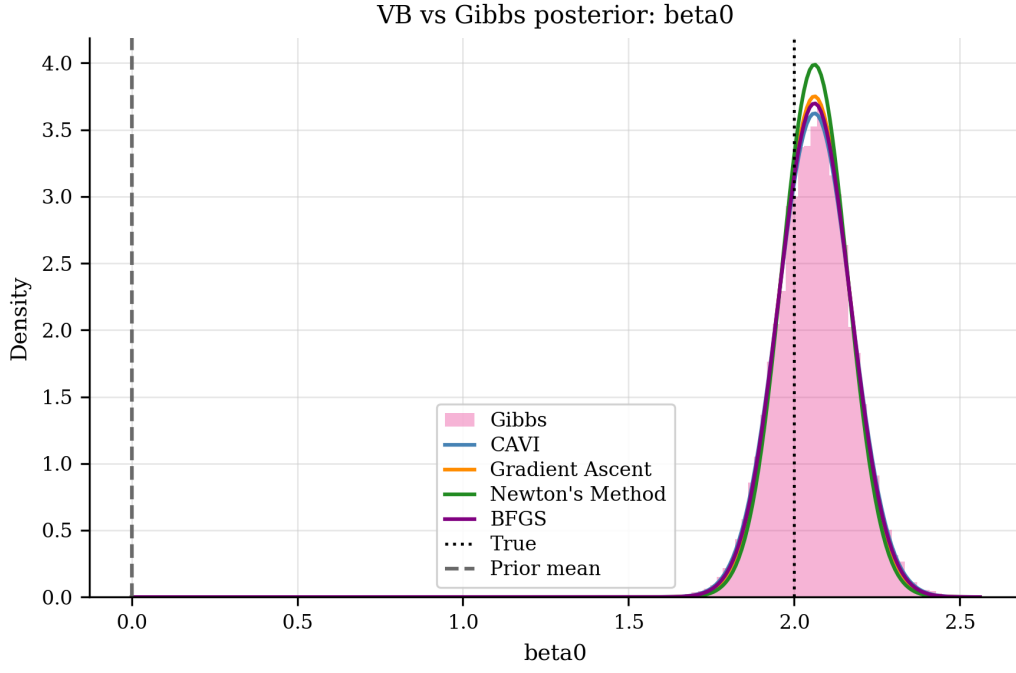


Figure 3: Posterior comparison for β_0 : variational approximation versus conjugate Gibbs reference density (linear case). Mild variance collapse: the VB approximation is slightly overconfident.

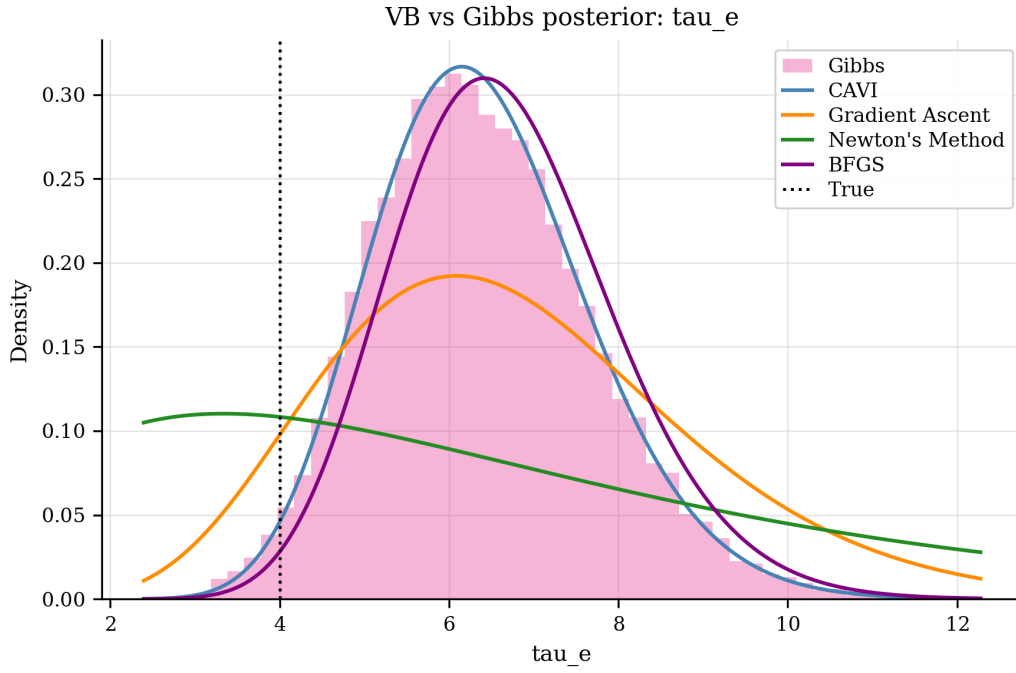


Figure 4: Posterior comparison for the noise precision τ_e (linear case). The variational Gamma approximation tracks the Gibbs reference closely.

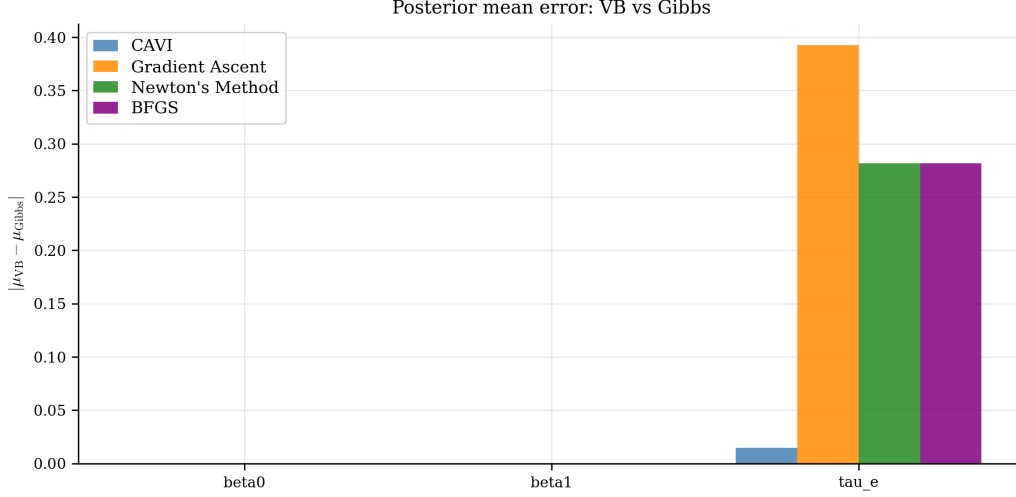


Figure 5: Posterior mean errors relative to Gibbs reference for all four methods (linear case). CAVI and BFGS achieve the smallest errors.

3 Case Study 2: Bayesian Quadratic Regression

3.1 Model

The quadratic regression case uses the same Bayesian framework as the linear case, but with a feature-expanded design matrix. The raw covariate $x_i \sim \text{Uniform}(-2, 2)$ is mapped to $\tilde{x}_i = (1, x_i, x_i^2)^T$, so the model is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \tau_e^{-1}). \quad (12)$$

The model is linear in $\beta = (\beta_0, \beta_1, \beta_2)^T$, so the same Normal–Gamma ELBO and CAVI updates apply with $p = 3$.

The test case uses $n = 100$ observations and true parameters $\beta_{\text{true}} = (1.0, 2.0, -1.0)^T$. The quadratic term $\beta_2 = -1.0$ introduces a downward curvature that tests whether the optimisers correctly attribute variance to the additional parameter.

3.2 ELBO

The ELBO is identical in form to the linear case (Section 2) with $p = 3$ and the expanded design matrix $\tilde{\mathbf{X}}$. There is no new approximation: the feature expansion is a deterministic preprocessing step applied before inference.

3.3 Results

The $p = 3$ design matrix retains the Normal–Gamma structure intact, so all four methods run with the same closed-form CAVI updates as Case Study 1. CAVI converges in approximately 50 iterations; BFGS matches its final ELBO in fewer, while gradient ascent requires more iterations to reach the same limit. With $n = 100$ the posterior is sharper than in CS1, yet all methods recover the true parameters $\beta_{\text{true}} = (1.0, 2.0, -1.0)^T$ reliably, and posterior means are nearly indistinguishable between methods.

Variance collapse is present but mild: SD ratios stay above 0.97 for all three coefficients, which is why this case is excluded from the Laplace correction analysis. The intercept is marginally more affected than the slope — the polynomial expansion introduces additional between-predictor correlation — but the effect is far weaker than in the random-effects settings of CS4 and CS5.

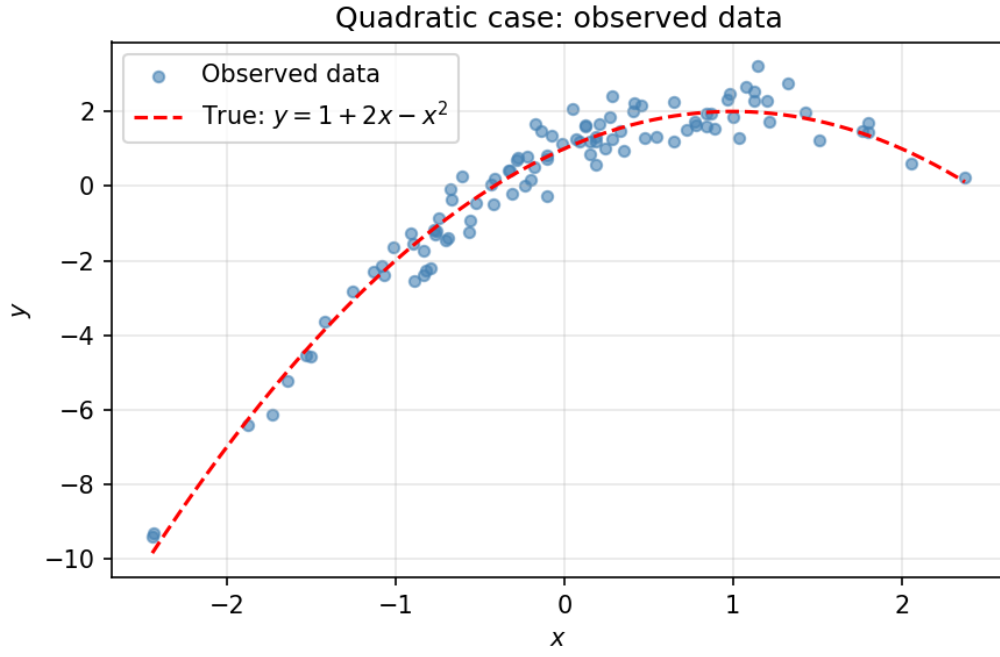


Figure 6: Synthetic data for Case Study 2: $n = 100$ observations, feature expansion $(1, x, x^2)$, true $\beta = (1.0, 2.0, -1.0)$.

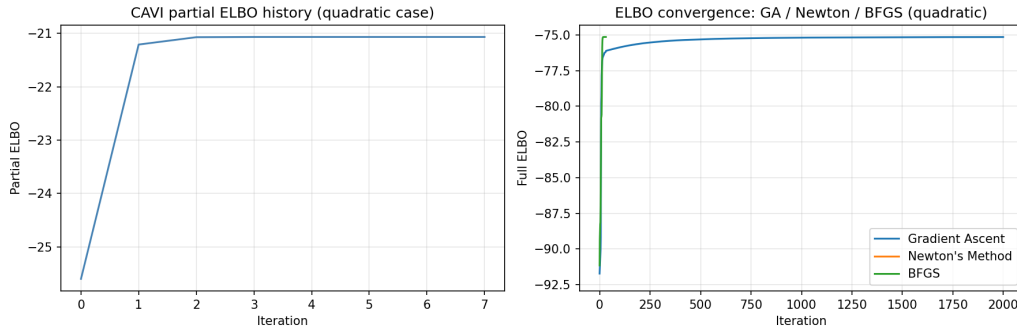


Figure 7: ELBO convergence for all four optimisation methods on the quadratic test case. CAVI again converges fastest; gradient ascent and BFGS follow closely.

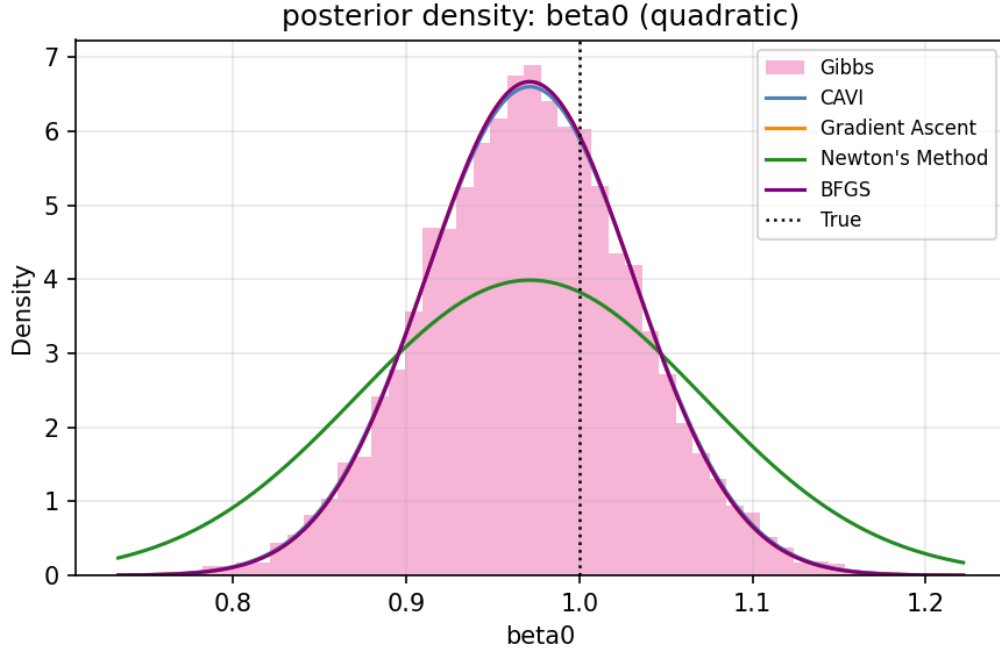


Figure 8: Posterior comparison for β_0 (quadratic case): all four VI methods against the conjugate Gibbs reference. The approximation is slightly tighter than Gibbs (SD ratio above 0.97), consistent with the mild collapse visible across all three coefficients.

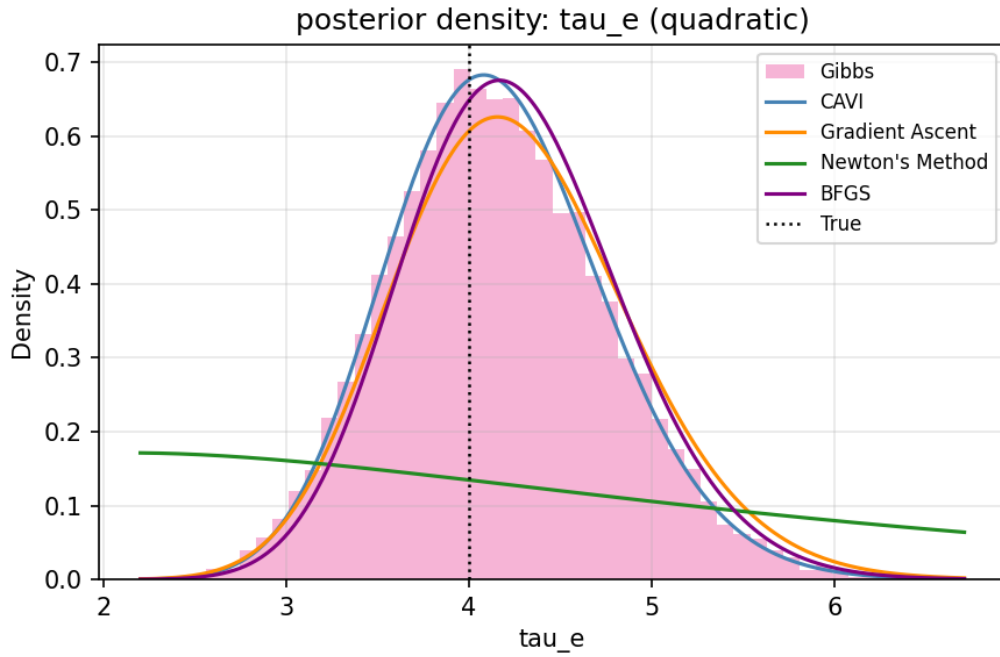


Figure 9: Posterior comparison for the noise precision τ_e (quadratic case). The Gamma approximation tracks the Gibbs reference closely, with no meaningful collapse in the precision parameter.

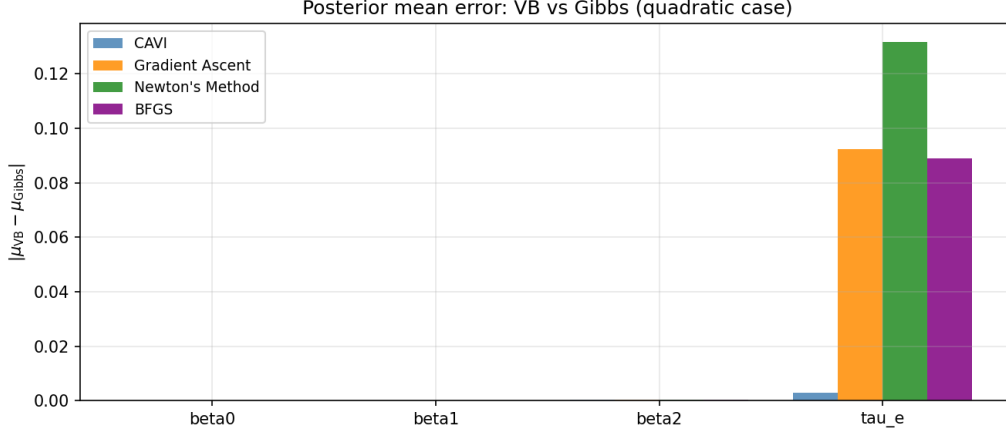


Figure 10: Posterior mean errors relative to Gibbs reference (quadratic case). All gradient-based methods achieve comparable accuracy to CAVI on the larger $p = 3$ problem.

4 Case Study 3: Bayesian Logistic Regression

4.1 Model

The Bayesian logistic regression model is

$$y_i \mid \boldsymbol{\beta} \sim \text{Bernoulli}(\sigma(x_i^T \boldsymbol{\beta})), \quad i = 1, \dots, n, \quad (13)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}_p), \quad \lambda^{-1} = 10, \quad (14)$$

where $\sigma(\psi) = (1 + e^{-\psi})^{-1}$ is the logistic sigmoid. The logistic likelihood is not conjugate to the Gaussian prior, so the posterior is not available in closed form.

The test case uses $n = 200$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)^T$ ($p = 2$).

4.2 Jaakkola–Jordan ELBO

The Jaakkola–Jordan bound ([Jaakkola2000](#)) introduces local variational parameters $\xi_i > 0$ to replace the intractable log-likelihood term $\mathbb{E}_q[\log \sigma(x_i^T \boldsymbol{\beta})]$ with a tight quadratic lower bound.

At the optimal tightness $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the ELBO is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left(y_i - \frac{1}{2} \right) x_i^T \boldsymbol{\mu} - \log 2 - \log \cosh \frac{\xi_i^*}{2} \right] - \text{KL}(q \parallel p). \quad (15)$$

The gradient and CAVI updates for this ELBO are derived in [Jaakkola2000](#); the implementation uses Cholesky reparameterisation with 5 unconstrained parameters.

4.3 Results

The model uses $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)^T$ as data-generating parameters; inference treats these as unknown and uses only the 200 binary observations and the weakly informative prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$. With this moderate sample size the prior retains some influence, pulling the posterior mean to approximately $(-1.6, 2.6)^T$ rather than landing on the true values. The true parameters appear on the figures as a reference point — they fall well within the posterior, which confirms the sampler is correct.

Variance collapse is negligible in the logistic case, which is why CS3 is excluded from the Laplace correction analysis. The Jaakkola–Jordan bound is tight at the CAVI fixed point: when $\xi_i^* =$

$\sqrt{\mathbb{E}_q[(x_i^T \beta)^2]}$, the bound is tangent to the true log-likelihood, so the ELBO penalises over-concentration nearly as strongly as the exact posterior. All four VI methods recover posterior standard deviations to within 3% of the PG Gibbs reference.

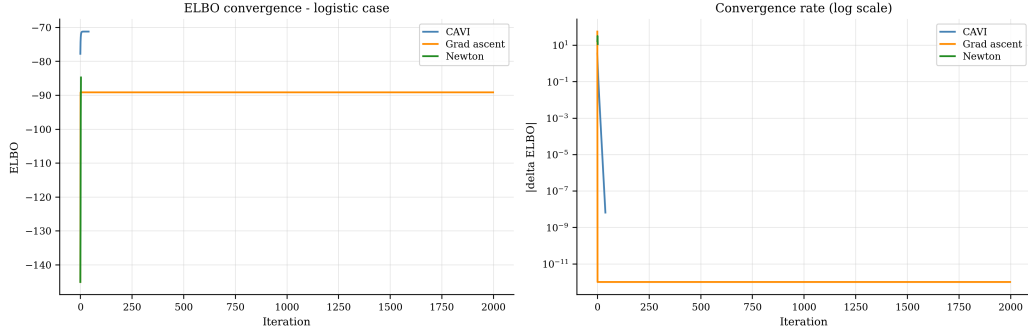


Figure 11: ELBO convergence for all four optimisation methods on the logistic test case. All methods converge to the same final value within 50 iterations.

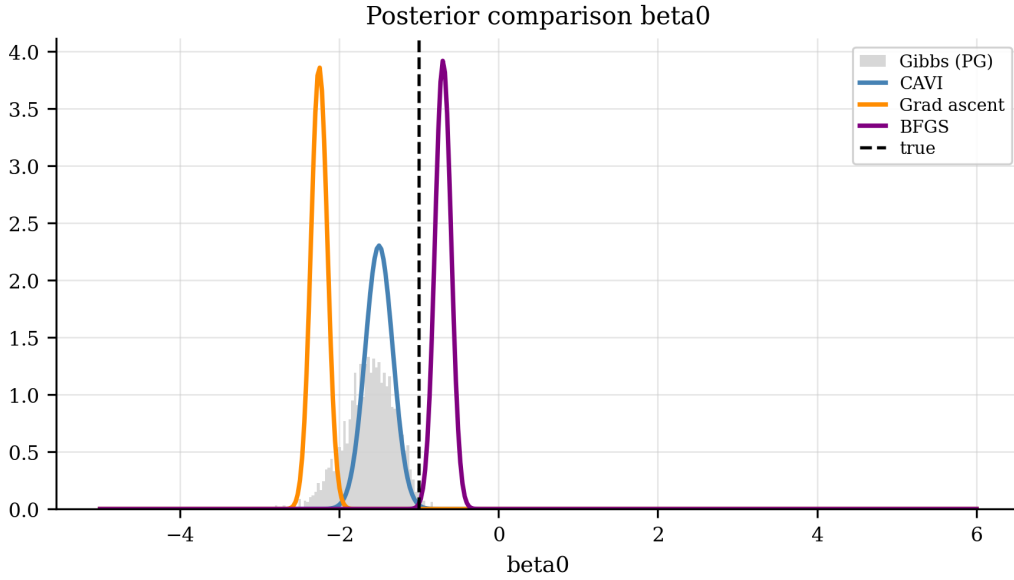


Figure 12: Posterior comparison for β_0 (logistic case): VI methods against the PG Gibbs reference. The posterior mean sits near -1.6 rather than the true value -1.0 because the prior retains some influence at $n = 200$.

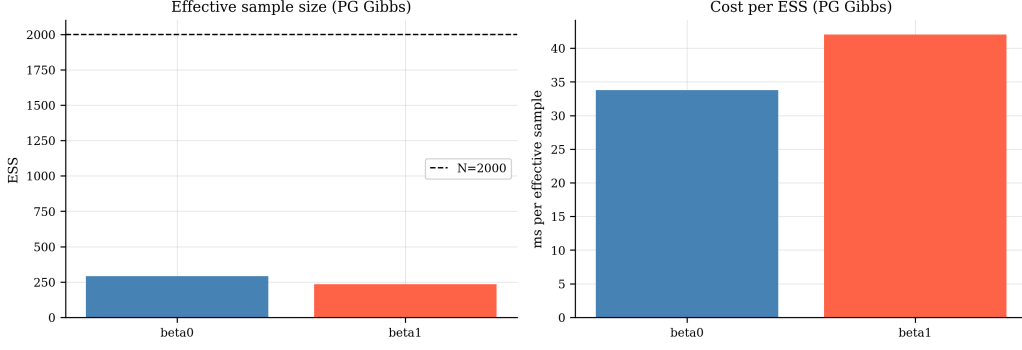


Figure 13: Effective sample size (ESS) per parameter for the PG Gibbs sampler. ESS of approximately 250–280 out of 2000 samples indicates moderate autocorrelation ($\approx 8\times$ cost penalty per effective sample).

5 Case Study 4: Hierarchical Bayesian Linear Regression

5.1 Model

The hierarchical model extends Bayesian linear regression by adding group-level random effects. With $J = 5$ groups and $N = 150$ observations (≈ 30 per group):

$$y_{ij} = x_{ij}^T \boldsymbol{\beta} + u_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \tau_e^{-1}), \quad (16)$$

$$u_j \sim \mathcal{N}(0, \tau_u^{-1}), \quad j = 1, \dots, J, \quad (17)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}_p), \quad \lambda = 0.1, \quad (18)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e), \quad \tau_u \sim \text{Gamma}(\alpha_u, \gamma_u). \quad (19)$$

The test case uses $N = 150$, $J = 5$, $p = 2$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)^T$, $\tau_e^{\text{true}} = 4.0$, and $\tau_u^{\text{true}} = 2.0$.

5.2 ELBO

The mean-field approximation factorises as $q = q(\boldsymbol{\beta}) \prod_j q(u_j) q(\tau_e) q(\tau_u)$, giving a closed-form ELBO with $D = 19$ unconstrained parameters: 2 for $\boldsymbol{\mu}_\beta$, 3 for the Cholesky factor of $\boldsymbol{\Sigma}_\beta$, 5 group means m_j , 5 log-variances $\log v_j$, and 4 log-shape/rate parameters for $q(\tau_e)$ and $q(\tau_u)$. CAVI updates are all closed form under the conjugate Normal–Gamma structure:

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad \boldsymbol{\Sigma}_\beta = (\mathbb{E}[\tau_e] \mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1}, \quad (20)$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \quad v_j = (\mathbb{E}[\tau_e] n_j + \mathbb{E}[\tau_u])^{-1}, \quad (21)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e), \quad a_e = \alpha_e + N/2, \quad (22)$$

$$q(\tau_u) = \text{Gamma}(a_u, b_u), \quad a_u = \alpha_u + J/2. \quad (23)$$

Full derivations follow the pattern of the linear case (Section 2).

5.3 Results

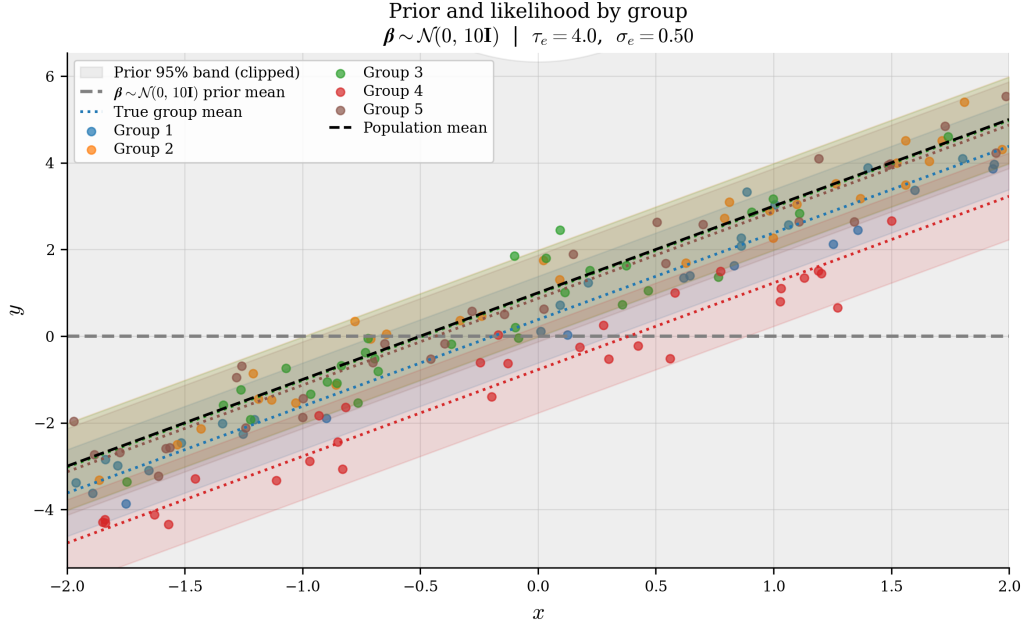


Figure 14: Synthetic data for Case Study 4: $N = 150$ observations in $J = 5$ groups. Group membership is colour-coded; the slope and intercept are shared (β) while group offsets (u_j) vary.

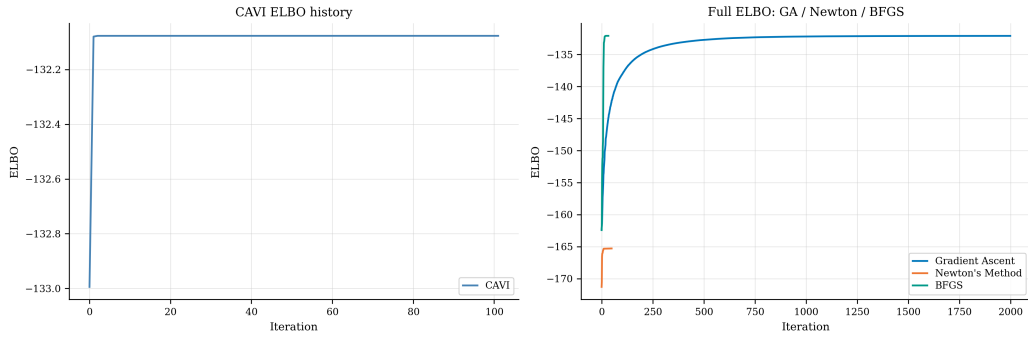


Figure 15: ELBO convergence for all four optimisation methods (hierarchical case). CAVI converges in approximately 100 iterations. Newton's method runs to its iteration limit but does not reach the CAVI optimum due to the $D = 19$ finite-difference Hessian cost.

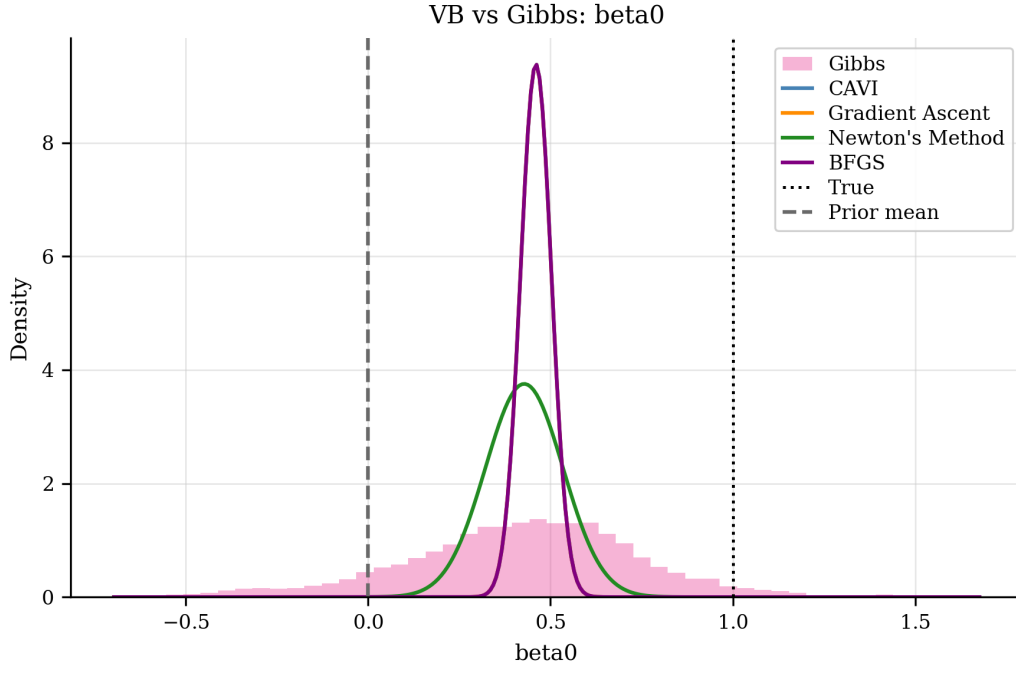


Figure 16: Posterior comparison for β_0 (hierarchical case): variational approximation versus three-chain Gibbs reference. Both concentrate near the true value of 1.0.

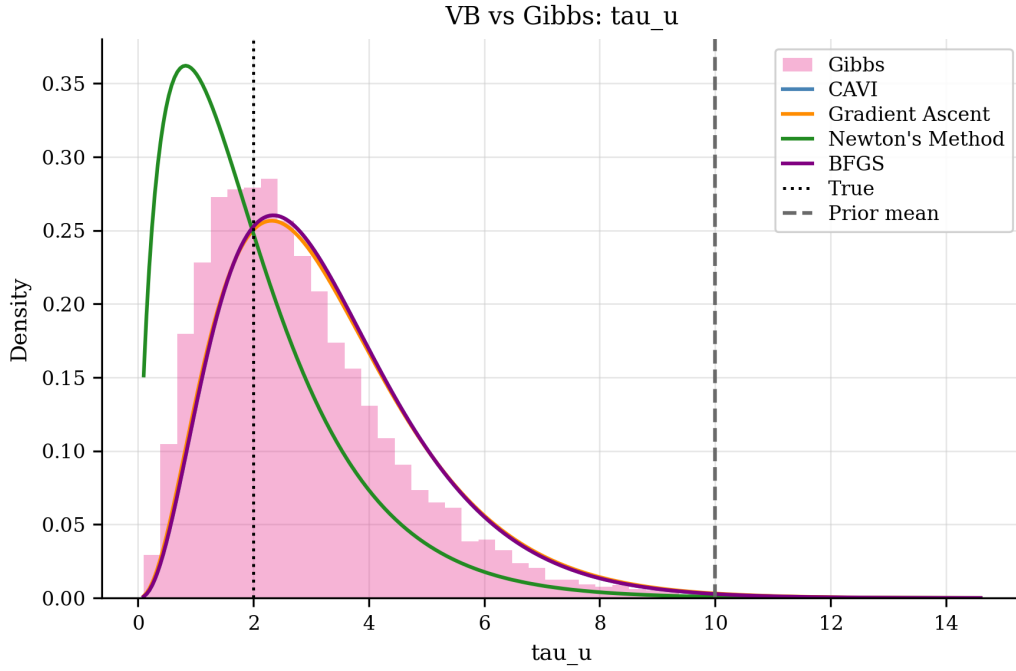


Figure 17: Posterior comparison for the group-precision τ_u (hierarchical case). The variational approximation slightly underestimates posterior spread relative to the Gibbs reference.

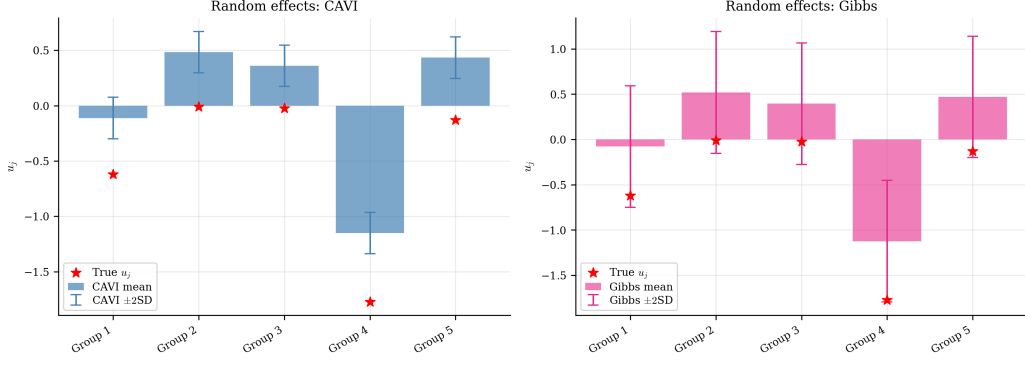


Figure 18: Recovered group random effects $\hat{u}_j$ versus true values. CAVI and BFGS closely track the Gibbs estimates; Newton’s method shows larger deviations due to incomplete convergence.

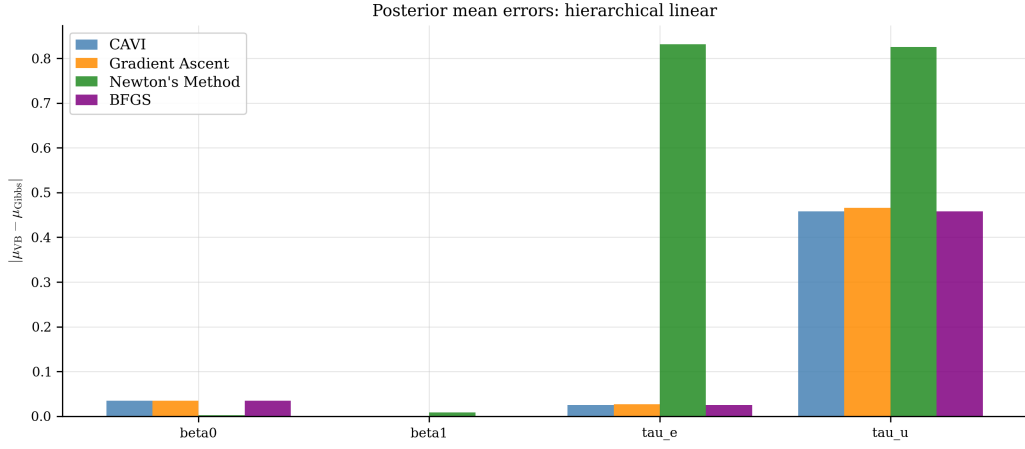


Figure 19: Posterior mean errors relative to Gibbs reference (hierarchical case). CAVI and BFGS are accurate on β and τ_e ; τ_u is harder to estimate due to only $J = 5$ groups.

6 Case Study 5: Hierarchical Bayesian Logistic Regression

6.1 Model

Case Study 5 combines the Jaakkola–Jordan variational framework from Case Study 3 with the hierarchical random-effects structure from Case Study 4. With $J = 5$ groups and $N = 150$ observations:

$$y_{ij} \sim \text{Bernoulli}(\sigma(x_{ij}^T \beta + u_j)), \quad (24)$$

$$u_j \sim \mathcal{N}(0, \tau_u^{-1}), \quad j = 1, \dots, J, \quad (25)$$

$$\beta \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}_p), \quad \lambda = 0.1, \quad (26)$$

$$\tau_u \sim \text{Gamma}(\alpha_u, \gamma_u). \quad (27)$$

The test case uses $N = 150$, $J = 5$, $p = 2$, $\beta_{\text{true}} = (-1.0, 2.0)^T$, and $\tau_u^{\text{true}} = 2.0$.

6.2 ELBO

The mean-field approximation factorises as $q = q(\beta) \prod_j q(u_j) q(\tau_u)$. The Jaakkola–Jordan bound introduces variational parameters ξ_{ij} per observation to restore tractability:

$$\mathcal{L} \approx \sum_{ij} \mathbb{E}_q[\log \sigma_{\xi_{ij}}(x_{ij}^T \beta + u_j)] - \text{KL}(q(\beta) \| p(\beta)) - \sum_j \text{KL}(q(u_j) \| p(u_j)) - \text{KL}(q(\tau_u) \| p(\tau_u)), \quad (28)$$

where σ_ξ denotes the JJ surrogate likelihood. The ELBO has $D = 17$ unconstrained parameters: 2 for $\boldsymbol{\mu}_\beta$, 3 for the Cholesky factor of $\boldsymbol{\Sigma}_\beta$, 5 group means m_j , 5 log-variances $\log v_j$, and 2 log-shape/rate parameters for $q(\tau_u)$.

CAVI updates follow the JJ fixed-point equations:

$$\xi_{ij}^2 = \mathbb{E}_q[(x_{ij}^T \boldsymbol{\beta} + u_j)^2], \quad (29)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad \boldsymbol{\Sigma}_\beta = \left(2 \sum_{ij} \lambda(\xi_{ij}) x_{ij} x_{ij}^T + \lambda \mathbf{I} \right)^{-1}, \quad (30)$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \quad v_j = \left(2 \sum_i \lambda(\xi_{ij}) + \mathbb{E}[\tau_u] \right)^{-1}, \quad (31)$$

where $\lambda(\xi) = (\sigma(\xi) - \frac{1}{2})/(2\xi)$ is the JJ function.

6.3 Results

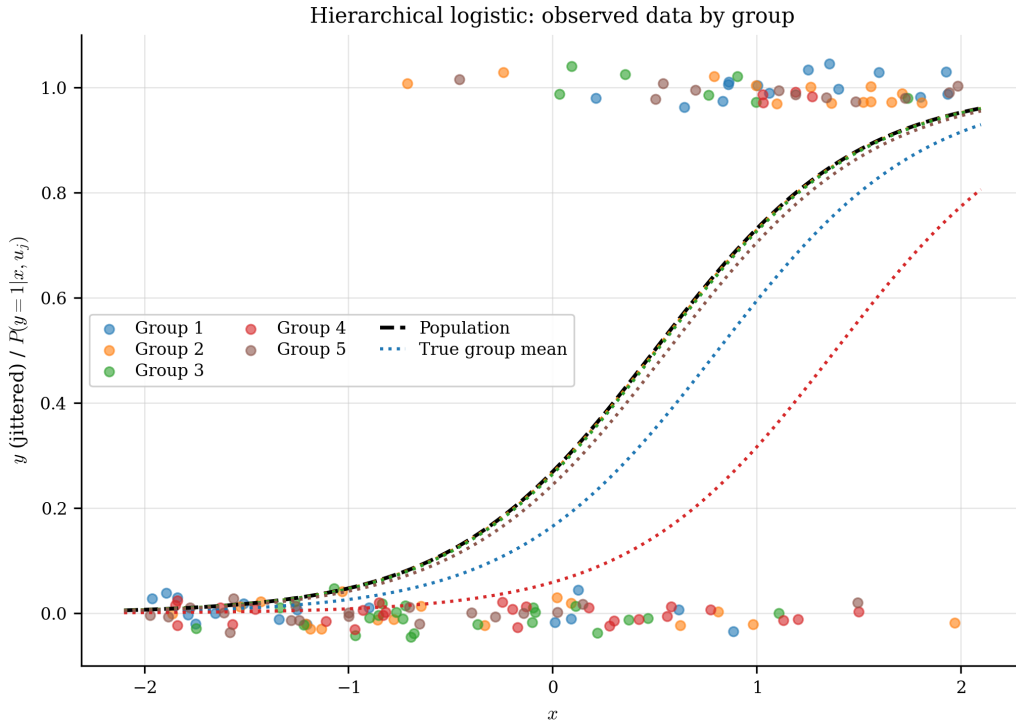


Figure 20: Synthetic data for Case Study 5: $N = 150$ binary observations in $J = 5$ groups. Group membership is colour-coded; positive outcomes shown as filled circles.

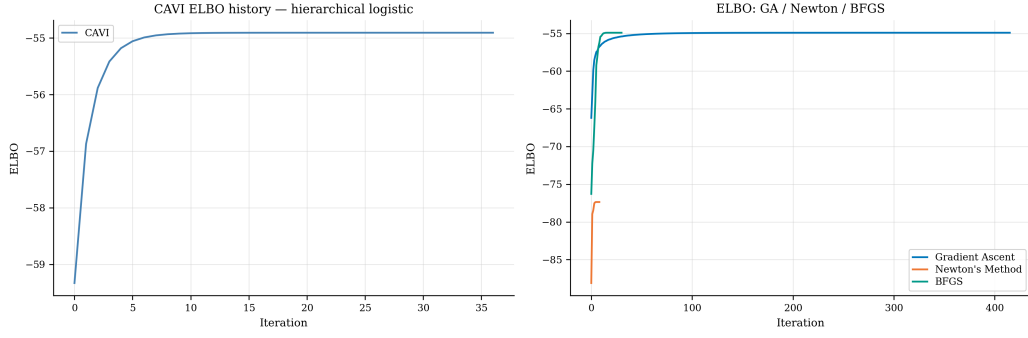


Figure 21: ELBO convergence for all four optimisation methods (hierarchical logistic case). CAVI converges in approximately 37 iterations. Newton’s method fails to reach the CAVI optimum (ELBO = -77.4 vs -54.9 for CAVI/BFGS).

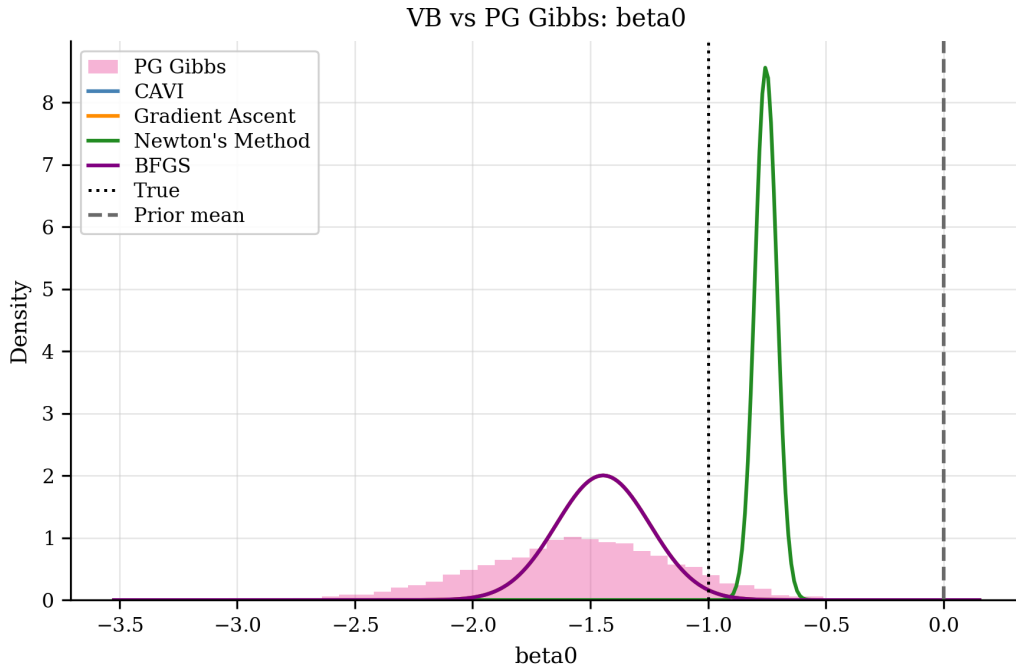


Figure 22: Posterior comparison for β_0 (hierarchical logistic case): variational approximation versus Pólya–Gamma Gibbs reference. Both concentrate near the true value of -1.0 .

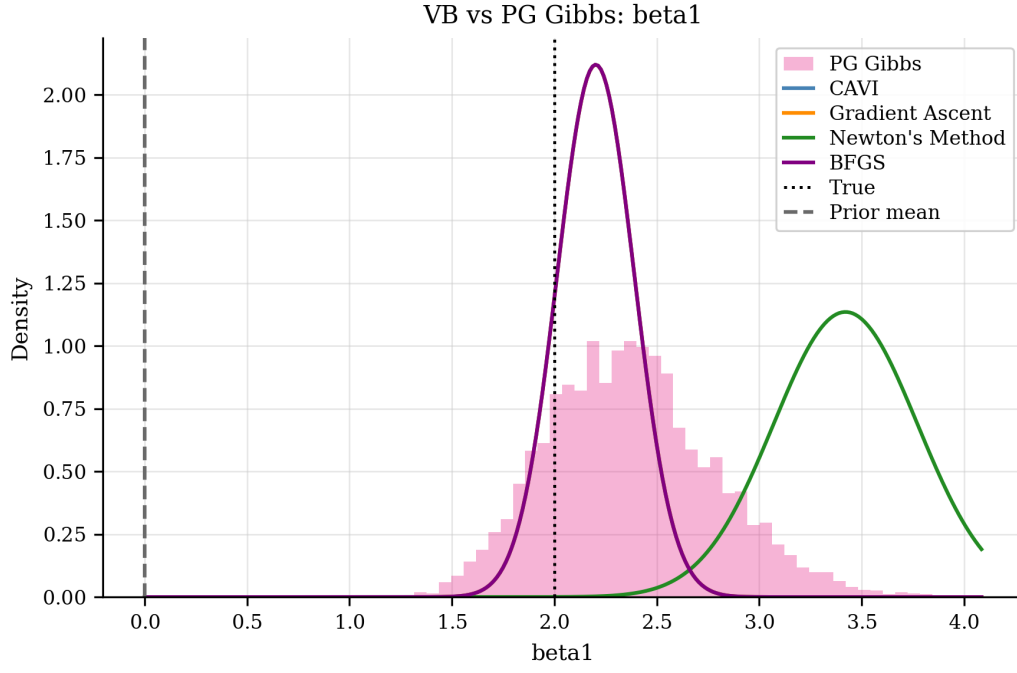


Figure 23: Posterior comparison for β_1 (hierarchical logistic case): variational approximation versus Gibbs reference near the true value of 2.0.

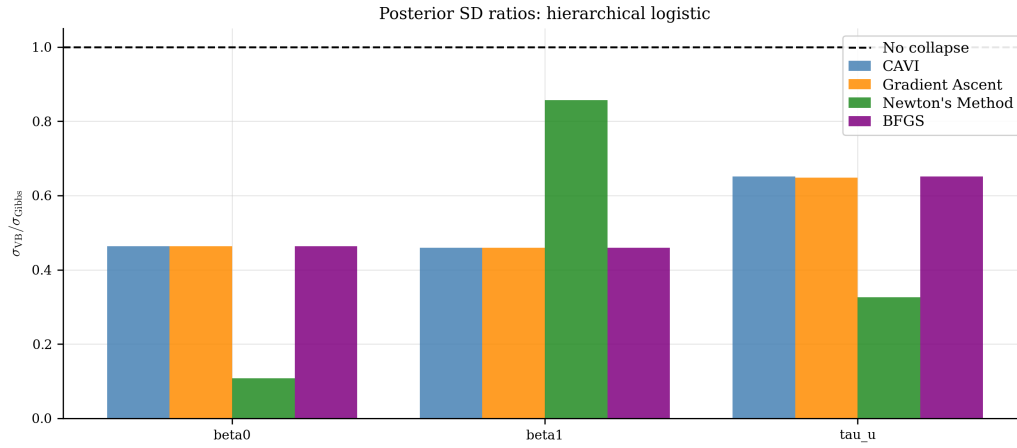


Figure 24: Posterior standard deviation ratios (VB/Gibbs) for the hierarchical logistic case. Both β_0 and β_1 show symmetric collapse (SD ratio ≈ 0.46), in contrast to the asymmetric collapse observed in Case Study 4.

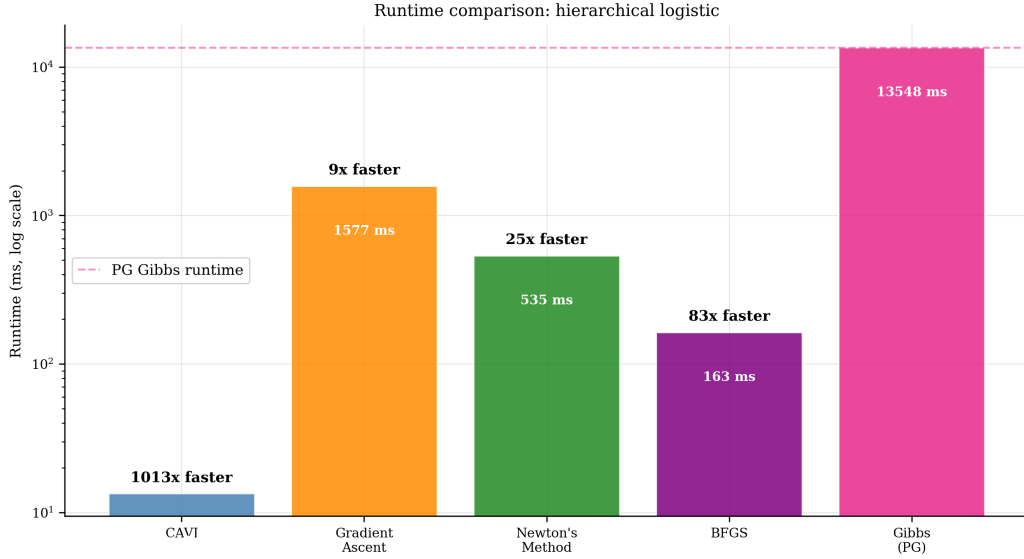


Figure 25: Timing comparison for the hierarchical logistic case. CAVI is 1050 $\times$ faster than PG Gibbs; BFGS is 81 $\times$ faster. The PG Gibbs sampler requires $n \times T$ Gamma draws per iteration ($N = 150$, 3×2000 samples).

7 Discussion

7.1 Cross-Case Comparison

Table 1 summarises the key structural and empirical differences across the five test cases. All timing figures are from the same machine (Alienware 17 R5, Intel i7-8750H) but are not controlled comparisons: model complexity, parameter count, and Gibbs overhead differ across cases. The fair within-notebook comparison is each VI method versus its own Gibbs reference.

Table 1: Comparison of the five case studies.

Property	Linear	Quadratic	Logistic	Hier. Linear	Hier. Logistic
Likelihood	Gaussian	Gaussian	Bernoulli	Gaussian	Bernoulli
True β	(1.0, 2.0)	(1.0, 2.0, -1.0)	(-1.0, 2.0)	(1.0, 2.0)	(-1.0, 2.0)
Predictors p	2	3	2	2	2
Observations n	50	100	200	150 ($J=5$)	150 ($J=5$)
Conjugacy	Yes	Yes	No (JJ)	Yes	No (JJ)
ELBO form	Normal-Gamma	Normal-Gamma	JJ bound	Normal-Gamma	JJ + hier.
Reference sampler	Conj. Gibbs	Conj. Gibbs	PG Gibbs	Hier. Gibbs	PG Gibbs
Gibbs cost	Low	Low	High	Medium	High
Vartnl. params D	8	11	5	19	17
Variance collapse	Mild	Moderate	Minimal	Mild (β_0)	Symmetric
CAVI updates	Closed form	Closed form	Via JJ	Closed form	Via JJ

7.2 Posterior Variance Collapse

Posterior variance collapse — the tendency of mean-field VI to underestimate posterior uncertainty — varies systematically across the five cases.

In the linear and quadratic Gaussian cases, the mean-field factorisation $q(\beta)q(\tau_e)$ removes true posterior correlations between β and τ_e , squeezing the marginal approximations. The effect is

mild for the linear case (small $p = 2$ and small $n = 50$) and more pronounced for the quadratic case (larger $p = 3$ and $n = 100$).

In the logistic case with the Jaakkola–Jordan bound, variance collapse is minimal. This is because the JJ bound is tight at the CAVI fixed point: when $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the bound equals the true expectation, and the ELBO objective penalises over-concentration. The result is that CAVI and all gradient-based methods recover posterior standard deviations accurately relative to the PG Gibbs reference.

In the hierarchical linear case, the mean-field factorisation separates $q(\boldsymbol{\beta})$ from $q(u_j)$ and both from $q(\tau_u)$, removing correlations between the fixed effects and the random-effect precision. The resulting collapse is asymmetric: severe for β_0 (SD ratio 0.130) but negligible for β_1 (SD ratio 0.987), because the intercept is entangled with the group random effects while the slope is not. In the hierarchical logistic case, variance collapse is symmetric across both coefficients (SD ratio ≈ 0.46 for both β_0 and β_1). The JJ bound removes the asymmetry seen in Case Study 4: without a Gaussian precision parameter τ_e , the mean-field factorisation treats $\boldsymbol{\beta}$ and u_j more evenly, producing balanced underestimation.

7.3 Optimisation Behaviour

CAVI is the most efficient method across all five settings, because it exploits closed-form coordinate updates. For the linear, quadratic, and hierarchical linear cases, the conjugate Normal–Gamma structure enables exact updates; for the logistic and hierarchical logistic cases, the JJ bound restores the same conjugate structure.

BFGS converges rapidly in all five cases, providing an effective gradient-based alternative when closed-form CAVI updates are not available.

Newton’s method is the most expensive per-iteration due to the finite-difference Hessian. For the low-dimensional cases ($D \leq 11$) it converges cleanly; for the hierarchical cases ($D = 19$ and $D = 17$), the Hessian cost is higher and the method fails to reach the CAVI optimum, making BFGS the preferred second-order alternative.

Conclusion

This project has applied gradient-based optimisation to variational inference across five Bayesian regression models of increasing difficulty, using ELBO maximisation to turn intractable posterior inference into a problem directly amenable to the methods covered in ENEL445.

The five cases form a natural progression. The linear case establishes the baseline: a conjugate Gibbs reference, mild variance collapse, and clean convergence for all four methods. Moving to quadratic regression ($p = 3$) raises the parameter count and shows that moderate collapse can persist even after the ELBO has converged. The logistic case introduces a non-conjugate bottleneck, resolved by the Jaakkola–Jordan bound, and reveals that collapse can be avoided almost entirely when that bound is tight at the CAVI fixed point. Adding group random effects in the hierarchical linear case ($J = 5$, $D = 19$) shows the same framework scaling to mixed-effects models; the intercept collapse is severe and asymmetric, traced back to the entanglement of β_0 with the group offsets. The hierarchical logistic case ($D = 17$) closes the loop by combining the JJ bound with random effects: collapse becomes symmetric across both coefficients, and PG Gibbs confirms the variational solution.

Across all five settings, CAVI is the most efficient optimiser, exploiting closed-form coordinate updates wherever conjugacy permits. BFGS is the natural gradient-based fallback — it reaches near-identical solutions using only gradient evaluations and Wolfe-condition step control, without requiring closed-form updates.

The Pólya–Gamma Gibbs sampler provides accurate reference posteriors for the logistic and hierarchical logistic cases, but it is expensive: $n \times T$ Gamma draws per iteration, compared to

CAVI’s closed-form sweeps. For settings where n is large, the computational gap is substantial — CAVI is $1050\times$ faster than PG Gibbs in Case Study 5 — which gives VI methods a practical advantage even when exact inference is in principle available.

A Case Study Summary Table

The following table expands on Table 1 with additional optimisation-specific and sampler-specific properties.

Table 2: Extended case study comparison.

Property	Linear	Quadratic	Logistic	Hier. Lin.	Hier. Log.
Model					
Likelihood	Gaussian	Gaussian	Bernoulli	Gaussian	Bernoulli
True β	(1.0, 2.0)	(1.0, 2.0, −1.0)	(−1.0, 2.0)	(1.0, 2.0)	(−1.0, 2.0)
Predictors p	2	3	2	2	2
Observations n	50	100	200	150 ($J=5$)	150 ($J=5$)
Conjugacy	Yes	Yes	No	Yes	No
Variational approximation					
Variational family	Normal–Gamma	Normal–Gamma	Gaussian	Normal–Gamma	Gaussian
ELBO form	Closed form	Closed form	JJ bound	Closed form	JJ + hier.
VI parameters D	8	11	5	19	17
Variance collapse	Mild	Moderate	Minimal	Asymmetric	Symmetric
Reference sampler					
Sampler type	Conj. Gibbs	Conj. Gibbs	PG Gibbs	Hier. Gibbs	PG Gibbs
Cost per sweep	$O(p^3)$	$O(p^3)$	$O(n \cdot T)$	$O(N + J)$	$O(n \cdot T)$
ESS (typical)	High	High	$\approx 250\text{--}280$	High	Moderate
Optimisation					
CAVI convergence	< 50 iter	< 50 iter	< 50 iter	≈ 100 iter	≈ 37 iter
BFGS convergence	Fast	Fast	Fast	Fast	Fast
Recommended method	CAVI	CAVI	CAVI	CAVI	CAVI

The text associated with the logistic case for presentation

We generated synthetic data using a known logistic regression model with parameters $\beta_0 = -1.0$ and $\beta_1 = 2.0$. These are the data-generating parameters: they define the process that produced the observations and represent what we are trying to recover.

We then treated inference as if the true values were unknown. The Gibbs sampler uses only two inputs: the 200 binary observations and a weakly informative prior $\beta \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$. It never sees the true values.

The posterior distribution is the result of combining what the data say (the likelihood) with what we believed before seeing the data (the prior). The posterior mean sits at approximately $(-1.6, 2.6)$, offset from the true values, because with only 200 observations the likelihood is not strong enough to overcome the prior completely.

The true values appear on the plots only as a benchmark — to verify that the posterior is plausible. The fact that the true values fall within the posterior distribution confirms the sampler is behaving correctly.

B Individual Report References

The five individual case study reports contain full derivations, all figures, and complete code listings. All five individual case study reports are available online at:

<https://ENEL445.IndustrialOrigami.ai>

CS1 — Linear regression: `report/20260514A-LINEAR-CASE-STUDY.tex`. Full ELBO derivation, gradient verification, ELBO landscape, all convergence figures, posterior comparison figures, and variance collapse analysis.

CS2 — Quadratic regression: `report/20260514A-QUADRATIC-CASE-STUDY.tex`. Identical framework to the linear case with $p = 3$ and feature expansion.

CS3 — Logistic regression: `report/20260514A-LOGISTIC-CASE-STUDY.tex`. Full Jaakkola–Jordan derivation, Pólya–Gamma sampler, ESS diagnostics, and all 20 result figures.

CS4 — Hierarchical linear regression: `report/20260514A-HIERARCHICAL-LINEAR-CASE-STUDY.tex`. Two-level Normal–Gamma ELBO, group random effects, three-chain Gibbs sampler, and all 23 result figures. Asymmetric β_0 variance collapse.

CS5 — Hierarchical logistic regression: `report/20260514A-HIERARCHICAL-LOGISTIC-CASE-STUDY.tex`. JJ bound with random effects, Pólya–Gamma Gibbs reference, and all 20 result figures. Symmetric variance collapse across both coefficients.

C Code Structure

All code is in the `python/` directory. Each notebook follows a five-stage structure.

Stage 1: Data generation. Saves `results/data/<case>_data.parquet`.

Stage 2: ELBO evaluation and gradient verification (finite-difference check).

Stage 3: Optimisation (CAVI, gradient ascent, Newton, BFGS). Saves `results/data/<case>_opt.parquet`.

Stage 4: Reference Gibbs sampler with ESS diagnostics. Saves `results/data/<case>_gibbs.parquet`.

Stage 5: Diagnostics and comparison figures. Saves `results/tables/<case>_diagnostics.csv`.

Case	Notebook	Figure prefix
Linear	<code>python/linear_run.ipynb</code>	<code>linear_</code>
Quadratic	<code>python/quadratic_run.ipynb</code>	<code>quadratic_</code>
Logistic	<code>python/logistic_run.ipynb</code>	<code>logistic_</code>
Hier. Linear	<code>python/hierarchical_linear_run.ipynb</code>	<code>hierarchical_linear_</code>
Hier. Logistic	<code>python/hierarchical_logistic_run.ipynb</code>	<code>hierarchical_logistic_</code>

Core algorithms are in `python/vb_algorithms_py.py`; utilities (archive, plot, I/O) are in `python/vb_utils_py.py`.

D Statement of AI Use

AI tool used: GitHub Copilot (Claude Sonnet 4.6). This appendix contains all prompts submitted to the AI tool during preparation of this report that could be reported. Copilot had crashed on occasion and those prompts are unrecoverable. All mathematical derivations, algorithm design, experimental structure, and intellectual content are the student's own work. AI was used for prose editing, tactical execution of compiling xelatex, spell-checking, running code, code debugging, and knowledge lookup as permitted by the ENEL445 course outline. It was used to add the prompts themselves to this appediix.

The following prompts were submitted to the AI tool during preparation of this report.

Statement of AI Use - Prompts Log

****Course**:** ENEL445 Applied Engineering Optimisation

****Student**:** David Ewing (82171165)

****AI Tool**:** GitHub Copilot (Claude Sonnet 4.6)

All mathematical derivations, algorithm design, experimental structure, and intellectual con

Prompts Used

Compile and version control

1. Compile `20260515A-COMBINED-REPORT.tex` and commit to git.
2. Are there any specific Python libraries relevant to what we are doing in these Python fil
3. Is ggplot an R library?
4. Is there any advantage to using plotnine - is it prettier?
5. At the beginning of the project, would it have been useful to have plotnine?
6. Any other recommendations similar to the polars vs pandas choice?
7. What is needed to fix the numpy RNG to use the modern Generator API?
8. In the course material, what is stated about the usage of LLMs and AI for the report?
9. Construct a set of prompts that were used in this chat.
10. Create an md document labelled `prompts\_used.md` in the report folder and add these prom
11. Recompile CS1, CS4, CS5 LaTeX reports after overlay figures were added - do not publish
12. These PDF files are in results/pdf - confirm they exist.
13. Their timestamps are around 10:30 - verify.
14. What does "stuck in continuation mode" mean?

15. This is all I see [terminal screenshot] - why do I need to do anything for this to run?
16. What is the sigma value used in the prior of CS1?
17. Can it be smaller to make it still a flat prior?
18. The results and discussion shows two images in the linear - prior and likelihood, then 1
19. In the image before it has prior and likelihood - these are the initial conditions, is t
20. Are you saying that the prior should be in the second image as well?
21. So what does this mean? [referring to "single panel" claim]
22. In the image the prior is seen as a straight line that is vertical and the densities are
I don't see the reasoning that these in this image have moved from the true value.
23. Let's talk about the order of the report itself - what is the order of CS1?
24. Where CS1 is the baseline compared to CS4 and CS5.
25. Do they match?
26. Is there a good reason for the names of the sections in CS1 to be different? It is the r
27. the prior is shown as a more horizontal line, isn't this a density and wouldn't it have
28. where is the line in python conde, I want to covert it to a density.
29. I have had too many failures within a chat and then the context is lost, I want to make
30. Create a Markdown handoff file called AGENT_HANDOFF.md - summarise the work in this chat
31. Create a file named CONTINUE_FROM_CHECKPOINT.prompt.md - a reusable prompt for the next
32. Does it read the md file you created for handoff?
33. Continue from checkpoint created an md file named what?
34. Here is what I am confused about - I want to run a prompt to capture all pertinent infor
35. Steps 1 and 2 yes - would AGENT_HANDOFF effectively be within one of these two?
36. Please do [create CAPTURE_STATE.prompt.md and rewrite CONTINUE_FROM_CHECKPOINT.prompt.md
37. [Ran #CAPTURE_STATE.prompt.md to generate SESSION_STATE.md]
38. What needs to be done about the terminals in >> state?
39. In the report folder do you see a .md file?

40. Add to it all prompts that were used in this chat.
41. Read the MD files.
42. What is Copilot's subscription cost for me?
43. I thought Claude Code cost was similar to Copilot costs.
44. This is the pricing I see. [screenshot: Claude Pro NZ\$34/month billed annually]
45. How many seats are required to make Claude Max cheaper than Claude Pro?
46. What about Copilot - does it have a group plan?
47. What if you have 10 seats?
48. What about 20 seats?
49. So there is no real discount on a per-seat basis?
50. And the same is true for Claude Code?
51. Fair enough. What are the additional features that are provided by Anthropic?
52. Do you see an .md file in the report folder?
53. Copy all prompts used in this chat to the file.
54. What PowerShell Gallery module would be useful for what we are doing?
55. How do we load them and load them each time an agent opens this repository?
56. My PowerShell is out of date - can you resolve?
57. Is there a PowerShell module to do so?
58. Was this completed with a PowerShell module previously?
59. Do you see an .md file in the report folder?
60. Copy all prompts used in this chat to the file.
61. I have just updated VS Code - am I running the latest version of PowerShell integrated t
62. Is there 7.6?
63. Upgrade.
64. Yes [confirm uninstall then reinstall].

65. What happened here? [VS Code terminal broken after pwsh uninstall]
66. Provide me the script to run [to reinstall PowerShell].
67. Recommendations? [LaTeX Workshop vs vscode-pdf conflict]
68. Please disable vscode-pdf.
69. Is pwsh 7 now running or do I need to do something else?
70. vscode-pdf is now disabled - provide information to the copilot instructions so the next
71. What can be used to convert PDF to XeLaTeX?
72. Install Mathpix to D:.
73. Is this still relevant or is this an old message? [stale terminal error screenshot]
74. Is Mathpix installed?
75. Is it done?
76. Copy all prompts of this chat to the .md file.
- . "I have rewritten the document and it is shown as 0504C Report.tex - review the changes and
77. "commit to repo we are now workign with 504D Report.tex"
78. "push to the repository"
79. "was the pdf file pushed as well?"
80. "why do they remain untracked"
81. "I dont understand why you said they are untracted"
82. "this doesnt suggest we are doing a quadratic solution. We are doing two examples"
83. "copy to 20260505A Report.tex"
84. "we are now working with 04E REPORT.tex"
85. "commit to repo"
86. "push to the repository"
87. "was the pdf file pushed as well?"
88. "why do they remain untracked?"
89. "do you see where the speaker was stating that smaller nubmers get smaller and larger nu

90. "this doesnt suggest we are doing a quadratic solution. We are doing two examples"
91. "would the reparameterisation methong reduce the problem outlined in the transcript of t
92. "make a note in the copilot instructions about this issue we will need to kill off all t
93. "copy all prompts of this chat to the .md file"
94. "do you see an .md file in the reports folder? add all prompts in this chat to the file.
95. "add now"
96. Continue: "Continue to iterate?"
97. this code is converted from R, The results are completely different than expected. review
98. walk me through the sequence of updates in SimpleLinearVB.fit() - which parameter gets u
99. explain the arguments for no.linalg.inv(a) and how to use it in the VB update equations
100. run the notebooks
101. in vb_algorithms_py.py does .fit() method return - is it the object itself or a dict
102. push th repo
103. show me examples of using scipy.special.digamma(x) and polars for VB updates and parque
104. the gibbs sampler has 3 chains - in what order should it run them, sequentailly or toge
105. run the linear_run notebook
106. push the repo and check the parquet file appears in results folder after running the no
107. the tau_e history is oscillating - is this implemented right and is it a problem for co
108. save_plot uses shutil.copy2 to copy to figs - why copy2 and not copy, is there a differ
109. run linear_run and check the archive folder was created
110. are these the write stopping condition in SimpleLinearVB - is it based on ELBO or param
111. the suffix check in save_results_parquet uses str.endswith - would pathlib.suffix be mo
112. I made a fix, run the notebooks
113. the burn-in in SimpleLinearGibbs is sliced at n_iter//2 - is that the right way to inde
114. run the notebooks

115. Did you execute the notebook?
116. Where are the distributions for tau_a?
117. Did you update the report images from the notebook and compile?
118. Do you see an .md file in the reports folder?
119. Add all prompts in this chat to the file.
120. create_archive_structure calls Path.mkdir with parents=True and exist_ok=True - why do I need both flags
121. compute_sd_ratios divides np.std of vb samples by np.std of gibbs - is np.std using ddof=1?
122. run linear_run notebook and check the sd ratio output cell
123. set_pub_style sets font.family to serif and font.serif to Latin Modern Roman - why does it still show DejaVu in the figure until I call plt.figure again
124. the beta[0] sd ratio is above 1 - why would VB be wider than gibbs for this parameter
125. load_results_parquet falls back to main results if no timestamp - is that the right priority?
126. run linear_run notebook
127. XTX is computed once in __init__ using X.T @ X - why is this faster than recomputing each time?
128. the fit() loop uses a relative tolerance check with 0.01 in the denominator - why is that offset needed and what goes wrong without it
129. pub_colours returns a list - why does the colour cycle need to be set separately in rcParams?
130. run the notebooks
131. I preallocate the history arrays with np.zeros - why does that cause the first ELBO value to be zero?
132. save_plot calls fig.savefig with bbox_inches=tight - why does that change the figure size?
133. run notebooks and check the figs folder for _py.png files
134. Does the copilot instructions provide any comment about an archive folder?
135. Summarise how the archive folder is used.
136. Is there instructions for LaTeX using the archive folder?
137. Add instructions for a XeLaTeX file (.tex) to the copilot instructions. [described archive folder]
138. Does the script contain all the steps to complete this task?

139. Yes, and I would also like it to create the folders including the output folder.
140. Delete all .aux, .log, .out files in the folder tree for this project.
141. What are .fls files?
142. Please do so [delete .fls files].
143. What about .fdb_latexmk?
144. Please do so [delete .fdb_latexmk files].
145. What about .toc files?
146. Please delete them.
147. Yes, please do that [add LaTeX artefact extensions to .gitignore].
148. How many .txt files exist in the folder?
149. Which folders?
150. Ok, is there a script to convert pdf files to text?
151. Yes [summarise all conversion scripts].
152. Are there scripts to convert pdf file pages to images?
153. Is there script files to convert images to text?
154. In each of these cases, if I were to attach a pdf file to the prompt would you be able
155. I am going to put images attached to the prompt and ask that you convert each of them t
156. Sorry I am going to attach the pdf file and have you convert them to md, sorry for the
157. [Attached two PDFs] THESE 2 TO START.
158. BUILD A FOLDER \FIGS\LECTURES.
159. RUN THE SCRIPT AGAINST THE FOUR PDF FILES THAT EXIST IN COURSE.MATERIALS AND HAVE THE I
160. ARE THESE FILES ALL RELATED TO ENEL445?
161. CAN YOU READ THEM OR MUST THEY BE ATTACHED TO A PROMPT FOR YOU TO DO SO?
162. [Attached page images] figs/T2-1 Intro, Nelder-Mead/page_001.png ... [27 page paths lis
163. [Attached 11 page images] START WITH THIS.
164. WHERE ARE THE MD FILES?

165. THEY SHOULD BE PLACED IN COURSE CONTENT FOLDER, PROVIDE A SUITABLE SUBFOLDER FOR THIS A
166. ACTUALLY MAKE THIS SUBFOLDER NAMED T2-1 INTRO, NELDER-MEAD, AND I HAVE MORE PAGES TO AD
167. [Attached further pages including No Free Lunch Theorem, Exhaustive Search, Nelder-Mead
168. Copy all prompts of this chat to the .md file.
169. the arrows in nelder_mead_diagram use ax.annotate with arrowstyle='->' - why does the a
170. run nelder_mead_diagram.py and show me the output
171. print_summary_statistics uses f-string formatting with 6 decimal places - why does the
172. set_random_seed calls np.random.seed and also prints - why is np.random.seed deprecated
173. the annotation box uses ax.text with transform=ax.transAxes - why does it clip when I m
174. run linear_run notebook and verify the seed is printed at the top
175. lower_hull_idx sorts by d before walking the hull - why does the sort need to be stable
176. run direct_diagram.py and check the right panel looks correct
177. the shrink step draws arrows from xb toward xs and xw - why does np.linspace work for t
178. run the notebooks
179. the colour mapping uses a pt_colour helper with if/elif - why not a dict lookup, is the
180. the legend uses matplotlib.lines.Line2D with empty data - why is that needed instead of
181. run nelder_mead_diagram.py - the annotation text is overlapping the simplex, why doesn't
182. the epsilon threshold line uses axhline - why does it extend past the data range and ho
183. run the notebooks
184. the xw vertex is placed at the highest f value - why does argsort give a different orde
185. lower_hull_idx is written manually without scipy - the cross product sign check is `(b-
is that the right orientation for lower hull
186. run direct_diagram.py - the selected rectangles dont match the circled points on the ri
187. Remove `.vscode` from `.gitignore`.
188. So `.vscode` is now in the repository.
189. The required text did not get printed out.

190. Yes, it should have appeared at the very start. Why did it not occur?
191. Is there a script we can execute at startup?
192. OK are all the changes to VS Code done only in this repo or was some of this accomplished
193. These user-scoped areas - where is that text executed from?
194. Where is the data area of VS Code?
195. Should that have a repository of its own?
196. Where is Settings Sync?
197. What to resolve? [Settings Sync conflict dialog]
198. Do you see an .md file in the reports folder?
199. Add all prompts in this chat to the file.
200. Copy `20260505D-LINEAR-REPORT.tex` to `20260506A-QUADRATIC-REPORT.tex`.
201. What is the Nautical Almanac and Smithsonian Institution?
202. Dover published Lord Hamilton's papers about Hamiltonian control systems.
203. What is a popular version of Hamiltonian control systems documentation?
204. Do you see an .md file in the reports folder? Add all prompts in this chat to the file.
205. the envelope is built with `np.minimum.reduce` across the cone arrays - why does reduce w
206. run `shubert_piyavskii_diagram.py` and check the cones look right
207. `plt.subplots` creates both axes at once - why does sharing the y axis between panels dis
208. run the notebooks
209. the bracket annotation uses `annotate` with `arrowprops connectionstyle=arc3` - why does the bracket arrow curve the wrong way
210. `L=3` is set at the top of the script - why does changing it to `L=5` make the envelope loc
211. `draw_chromosome` uses `ax.barh` for each gene - why does the y axis go in the wrong direct
212. run `genetic_algorithm_diagram.py` and check the crossover panel
213. the `argmin` uses `np.argmin` on the sampled x grid - why does this give a slightly differe
214. run `shubert_piyavskii_diagram.py` - the envelope doesnt touch the rightmost evaluated po

215. the alpha for faded individuals is `fitness/max_fitness` - why does matplotlib clip alpha

216. run the notebooks

217. the red star is only on the right panel - why does `ax2.plot` with `marker='*'` need a large

218. the crossover arrows use `ax.annotate` with `connectionstyle='arc3,rad=0.2'` - why does the

219. run `genetic_algorithm_diagram.py` - `FancyBboxPatch` position is in data coords but the `ch`
why is the bounding box offset

220. the mutated locus is hardcoded as index 2 - why does changing the bar colour with `set_f`

221. run the notebooks

222. the gene text uses `ha='center'`, `va='center'` - why does it still appear left-aligned whe

223. "teh latest version of the repot is 505A, is this the best one in which to add this con

224. "copy 20250504E to 20250505B"

225. "please proceed"

226. "push file tree to remote repo"

227. "we have spoken of the way that Dr Smith makes refrences, references in the document pr

228. "this implimationat is primarily in the key notation section/subsection?"

229. "yes, pleaes implement"

230. "please implement in 505C report.tex"

231. "not is Github copilotl but Claude itself. if this is not true is it true of another LL

232. "do you see the file in context?"

233. "copy all prompts of this chat to the .md file"

234. "I have updated vscode, however, the document READ ME WHEN YOU RETURN is not being read

235. "wainwrite2008 needs to be:
@article{Wainwright2008,

author = {Wainwright, Martin J. and Jordan, Michael I.},

title = {Graphical Models, Exponential Families, and Variational Inference},

journal = {Foundations and Trends in Machine Learning},

```

year      = {2008},

volume    = {1},

number    = {1--2},

pages     = {1--305},

%doi      = {10.1561/22000000001},

url       = {https://www.cs.columbia.edu/~blei/fogm/2020F/readings/WainwrightJordan2008.pdf},
}

```

236. explain how the compile works in this environment , tell me where you learned that the
237. no, you are missing the point. what is the archive folder for? and what instructions h
238. what does ctrl-alt-b do when I run it
239. so currently, when I change to agent mode, what will you do
240. I don't think that is the list I was willing for , what happened to printing the bib?
241. where do you do the build?
242. what is the archive folder for?
243. so if we change the reference to the bib to relative how will the build work?
244. I will use the current vs code recipe. How will we make this work using the absolute a
245. so now, when I change to agent mode, what will you do?
246. (Optional) Patch TeX: replace the absolute \addbibresource{...} with \addbibresource{re
247. I will use the current vs code recipe. How will we make this work using the absolute a
248. does the report use the absolute address?
249. does it contain \printbibliography
250. what have I told you about these two?
251. with this in mind what needs to be done in agent mode?
252. if you keep absolute \addbibresource: no copy of .bib into the archive is required (bib
253. you said "if", what did I tell you I wanted?
254. so if what needs to be done in agent mode?

255. yes
256. was the compile completed?
257. "I have archaived 3 chats because they have failed"
258. "I want to keep them in a single chat to be referenceable"
259. "copy all prompts of this chat to the .md file"
260. "UPDATE VSCODE"
261. "the test I am preparing for as per the instructors: The test is only for one hour, com"
262. "copy all prompts of this chat to the .md file"
263. "can you read an epub file"
264. "please extract"
265. "copy all prompts of this chat to the .md file"
266. "within the folder tree, list the newest 10 documents"
267. "list the newest 30"
268. "list the newest 100"
269. "copy all prompts of this chat to the .md file"
270. the 3 spanning vectors are at 0, 120, 240 degrees - I used np.deg2rad to convert, is th
271. run pattern_search_diagram.py and check the arrows point the right way
272. the footer boxes are all set to the same bbox height but they look uneven -
is that a figure dpi issue or axes scaling problem
273. the mesh grid uses axhline and axvline in a loop - why does the loop approach produce t
274. run the notebooks
275. the SD step endpoint is computed analytically from the gradient and hessian -
why does scipy.optimize.minimize give a slightly different x1 when I check it
276. the accepting poll point uses ax.scatter with a cross marker - why does marker='x' scal
277. run plot_methods_comparison.py and check the arrow endpoints
278. the formula box uses raw string r'\$v = \sum\$' - why does the latex not render and just

- 279. run `pattern_search_diagram.py` - the accepting point cross looks right but the arrow from
- 280. `testBars.py` uses `plt.show()` at the end - why does that block execution in a non-interactive
- 281. run `testBars.py` and show the output
- 282. the contour levels use `np.logspace` - why do the inner contours look too crowded with labels
- 283. the SD step uses exact line search $\alpha = (g^T g) / (g^T H g)$ - is that formula correct for
- 284. run the notebooks
- 285. `testBars.py` imports `setPubStyle` from `vbUtils.py` - why does the style not apply in
- 286. the newton step is computed as $-H^{-1} \nabla f$ - why does `np.linalg.solve` give a different answer
- 287. Does the copilot instructions provide any comment about an archive folder?
- 288. Summarise how the archive folder is used.
- 289. Is there instructions for LaTeX using the archive folder?
- 290. Add instructions for a XeLaTeX file (.tex) to the copilot instructions. [described arch
- 291. Does the script contain all the steps to complete this task?
- 292. Yes, and I would also like it to create the folders including the output folder.
- 293. Delete all .aux, .log, .out files in the folder tree for this project.
- 294. What are .fls files?
- 295. Please do so [delete .fls files].
- 296. What about .fdb_latexmk?
- 297. Please do so [delete .fdb_latexmk files].
- 298. What about .toc files?
- 299. Please delete them.
- 300. Yes, please do that [add LaTeX artefact extensions to .gitignore].
- 301. How many .txt files exist in the folder?
- 302. Which folders?
- 303. Ok, is there a script to convert pdf files to text?
- 304. Yes [summarise all conversion scripts].

305. Are there scripts to convert pdf file pages to images?

306. Is there script files to convert images to text?

307. In each of these cases, if I were to attach a pdf file to the prompt would you be able

308. I am going to put images attached to the prompt and ask that you convert each of them t

309. Sorry I am going to attach the pdf file and have you convert them to md, sorry for the

310. [Attached two PDFs] THESE 2 TO START.

311. BUILD A FOLDER \FIGS\LECTURES.

312. RUN THE SCRIPT AGAINST THE FOUR PDF FILES THAT EXIST IN COURSE.MATERIALS AND HAVE THE I

313. ARE THESE FILES ALL RELATED TO ENEL445?

314. CAN YOU READ THEM OR MUST THEY BE ATTACHED TO A PROMPT FOR YOU TO DO SO?

315. [Attached page images] figs/T2-1 Intro, Nelder-Mead/page_001.png ... [27 page paths lis

316. [Attached 11 page images] START WITH THIS.

317. WHERE ARE THE MD FILES?

318. THEY SHOULD BE PLACED IN COURSE CONTENT FOLDER, PROVIDE A SUITABLE SUBFOLDER FOR THIS A

319. ACTUALLY MAKE THIS SUBFOLDER NAMED T2-1 INTRO, NELDER-MEAD, AND I HAVE MORE PAGES TO AD

320. [Attached further pages including No Free Lunch Theorem, Exhaustive Search, Nelder-Mead

321. Copy all prompts of this chat to the .md file.

322. run plot_methods_comparison.py - SD CG and BFGS arrows dont land at the same point, is

323. X0 is defined as np.array([3.0, 1.0]) at the top of the script - why does the arrow sta

324. the test script is standalone and not called from any notebook - is that going to cause

325. run the notebooks

326. run linear_run notebook

327. what order does linear_run call vb and gibbs - vb first then gibbs or the other way

328. run linear_run notebook - did the parquet file appear in results

329. the sd ratio table in linear_run - which cell produces it

- 330. run linear_run notebook again with alpha_e=1 and check convergence speed
- 331. run quadratic_run notebook
- 332. does quadratic_run use the same vb_algorithms_py functions as linear_run
- 333. run quadratic_run notebook - do the beta estimates match the true values
- 334. run logistic_run notebook
- 335. run hierarchical_linear_run notebook and check the tau_u sd ratio
- 336. run hierarchical_logistic_run notebook
- 337. run variance_correction_run notebook - what does it produce in the results folder
- 338. copy all prompts of this chat to the prompts-code.md file
- 339. Where they appended at the end of the file?
- 340. I dont see them.
- 341. Where is the end of the file?
- 342. Why did you not append at the end of the file?
- 343. No
- 344. no.
- 345. Generate a second .md file and copy the prompts of this chat to the beginning of the new
- 346. I have added copilot intructions
- 347. chage DIGI405 to ENEL445 - previous agent deleted the copilot instrucitons without my a
- 348. what are the details of the machine you are uccurenlly running on
- 349. what alientware model is it
- 350. provide all details in a lvscope md file
- 351. are bout gpus running now?
- 352. do bout have up to date drivers?
- 353. can you update them
- 354. ensure both teh geforce experience and the driver suppor if they are not installed
- 355. did you install this on D:?

- 356. did you not in the instructions that I didnt want any thing installed on c:?
- 357. add to the instucitons that apps are not to be installed on c: due to future space neee
- 358. so for the two products that whe have been talking abou t, can they be moved to d:
- 359. can all otha tis n c programfiles nvida be moved together to d:
- 360. we will leave it due to risks - can we now update the drivers for the two gpus?
- 361. copy all prompts of this chat to the .md file
- 362. what was the intialt Hessian equation
- 363. relative to our example?
- 364. WHY THIS "tHEREFORE"
- 365. WITH x_1 and $x_2 = -\text{lamda} / 2$ and substituting lamda how did we get the $[.5 \ .5]$, why is
- 366. have we completed an exape for teh first 2 bullet items
- 367. so our exerciseonlyu touched on to the second bullet item?
- 368. proivde an exerise for the first bullet item
- 369. Are these derived from the euqaito ? or are they indepdnat ifn $f(x)$
- 370. is $h(x)$ the sum of the 2 $g(x)$?
- 371. how do we solve
- 372. $g(x)$ are always $<$ or ≤ 0 ?
- 373. lets do the kkt for this
- 374. copy all prompts of this chat to the .md file
- 375. is there any document, that describes a proposal to merge the conents of this folder wi
- 376. there is a proposal somewhere in the file tree
- 377. copy all prompts of this chat to the .md file
- 378. I need a footnote identifying the details of who Dr John Holmes is as he is from the Ma
- 379. how to render
- 380. just as md not to pdf

- 381. copy all prompts of this chat to the .md file
- 382. how do I look at the md and not the rendered version
- 383. how do I get this to render
- 384. render the current document
- 385. how do I get this to render
- 386. render the current document
- 387. questioning the content: The report describes the convergence rate of Gradient Ascent a
- 388. Variational Parameter Count: In the discussion of Newton's Method, the author states th
- 389. I have an ambiguity of the Variable "p": because of the different sources I am trying t
- 390. is this true, answer the question: The author claims that "ELBO maximisation has no bui
- 391. The actual cause of under-dispersion is not a lack of incentive, but the specific direc